Class Imbalance Mitigation for TCGA-UT

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Abstract

Class imbalance in computational pathology depends on the input regime: labeled image patches support ordinary classification, whereas whole-slide image (WSI) diagnosis is commonly modeled as weakly supervised learning over patch-feature bags. This report compares representative class-imbalance mitigation categories on TCGA-UT, a moderately imbalanced (head-to-tail ≈28:1) histopathology dataset representative of real-world clinical class imbalance, using frozen Virchow2 representations for patch classification and WSI-bag classification on shared case-disjoint splits and three seeds. Each method is tuned over one imbalance-specific control selected on validation; held-out test results then rank the methods. Center-focused affinity loss (CFAL) reaches the highest patch-level mean macro F1; Odd-K-Out (OKO) set learning reaches the highest balanced accuracy and the best raw NLL and Brier among patch methods, although no patch method is uniformly best on every calibration metric; and CE plus soft-F1 with balanced sampling also improves both discrimination metrics, but by narrower margins. ProGAN augmentation and focal loss remain near the CE baseline. On WSI bags, RankMix has the highest mean balanced accuracy and MDE-MIL the highest mean macro F1; before temperature scaling MDE-MIL also has the best probability scores (lowest NLL, Brier, and ECE), but after a shared post-hoc step no WSI method is uniformly best on all calibration metrics, and the WSI gaps are small relative to seed-to-seed variation. Two lessons follow. First, once a strong frozen pathology encoder is fixed, patch-level imbalance gains come from reshaping the decision head or objective—as in CFAL's prototype affinity loss or OKO's set-level logit aggregation—rather than from resampling, difficulty reweighting, or synthetic-image augmentation, which barely move the CE baseline. Second, the most effective mechanism differs between the patch and WSI-bag regimes, so imbalance mitigation should be selected and evaluated in its native input regime alongside classwise recall and calibration, not treated as a universally transferable recipe.

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1 Introduction

Class imbalance changes what a pathology classifier is rewarded for learning. When common tumor types dominate training data, high aggregate accuracy can coexist with unreliable rare-class recall, especially if evaluation emphasizes prevalence-sensitive metrics. Long-tailed learning and medical-image evaluation literature therefore motivate classwise recall, macro F1, balanced accuracy, and calibration-aware probability metrics as primary outcomes in imbalanced settings (Saini and Susan, 2023; Zhang et al., 2023; Mosquera et al., 2024).

Imbalance methods often assume a specific prediction unit. Image-level methods use labeled images, image embeddings, or directly generated image samples, whereas WSI-level methods use weak labels over bags of instances together with attention, instance ranking, or bag-level representation learning. This report therefore evaluates each method family in the input form it is designed to consume.

TCGA-UT provides the inputs needed for both parts of the benchmark. The dataset includes labeled histopathology patches cropped from pathologist-selected tumor regions and slide identities that can be grouped into WSI-level bags. Patch labels are class labels, not dense pixel masks or cell-level annotations. This report uses that supervision directly: patch-level methods are evaluated on controlled patch classification with frozen Virchow2 embeddings, while MIL methods are evaluated on slide-level feature bags built from the same foundation-model family.

The purpose is not to introduce a new mitigation method or to reproduce every cited method in all implementation detail. Instead, the report compares the effectiveness of major class-imbalance mitigation categories on a current pathology foundation-model representation for both controlled patches and WSI feature bags. It implements representative data-level, loss-level, hybrid, synthetic-generation, and MIL-specific interventions under fixed splits and frozen Virchow2 features, tunes one imbalance-specific control per method on validation, and reports discrimination, classwise recall, and calibration on the held-out test split. Method fidelity is defined by implementing each method's defining imbalance mechanism while holding the backbone family, split, input budget, and evaluation code fixed within each benchmark.

2 Dataset

TCGA-UT is derived from diagnostic *whole-slide images* (WSIs) in The Cancer Genome Atlas (TCGA), a pan-cancer resource that links molecular and clinical cancer data with large image collections (Tomczak et al., 2015). The TCGA-UT release turns this slide archive into a computational-pathology image dataset by extracting hematoxylin and eosin (H&E) tumor *patches* from pathologist-selected *uniform tumor regions*. It contains 1,608,060 RGB patches of size 256×256 pixels from 8,736 diagnostic slides and 32 *cancer-type labels*. The patch labels are slide- or cancer-type labels, not dense tissue masks or cell-level annotations (Komura et al., 2022).

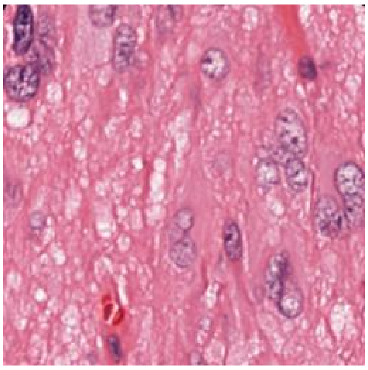
This structure makes TCGA-UT well suited for an imbalance study because the imbalance is native to the dataset rather than imposed only after sampling. The largest class is glioblastoma multiforme (GBM; 792 slides), while the smallest class is diffuse large B-cell lymphoma (DLBC; 28 slides). The resulting slide-level imbalance ratio is 28.3. At this ratio, TCGA-UT exhibits moderate, realistic class imbalance rather than an extreme long tail, which makes it representative of the class imbalance common in real-world clinical cohorts. As a consequence, the dataset exposes the practical tension that motivates this study: methods must learn from many visually related cancer categories while preserving performance on classes with far fewer slide-level examples.

Property	Value
Cancer-type labels	32
Diagnostic slides	8,736
TCGA-UT patches	1,608,060 patches of 256×256 pixels
Largest class	Glioblastoma multiforme (GBM), 792 slides
Smallest class	Diffuse large B-cell lymphoma (DLBC), 28 slides
Slide-level imbalance ratio	28.3
Shared split size per seed	5,025 train, 1,075 validation, 1,075 test cases; slide counts vary by seed
Leakage-control status	Case-disjoint by TCGA participant barcode; slide-disjoint follows from case assignment (Section 2.2)
Controlled patch input	Resolution level 0; up to 30 deterministic patches per slide; one frozen Virchow2 embedding per patch (Section 2.2)
WSI-bag input	Up to 30 deterministic frozen Virchow2 feature instances from the same slide identities

Table 1: Dataset properties used by the shared patch and WSI-bag benchmarks.

2.1 Data Units and Terminology

Several data units are involved in the benchmark, and the distinction matters for method comparison. A *slide* or *WSI* is one diagnostic histopathology image associated with one cancer-type label. A *case* or *patient* is the TCGA participant identified by the first three fields of a TCGA slide barcode, for example `TCGA-OR-A5J1`. A *tumor region* is a pathologist-selected area of a slide from which TCGA-UT crops groups of patches. A *patch* is one 256×256 RGB crop selected by the controlled manifest. A *patch embedding* is the frozen Virchow2 vector computed from that crop and consumed by the patch-level linear classifier. A *feature chunk* is a saved frozen-feature representation associated with a slide or slide subset. A *WSI bag* is the collection of frozen patch-feature instances used by the attention-MIL model for one slide-level prediction. *Head*, *body*, and *tail* classes refer to fixed support tiers derived from the full TCGA-UT slide distribution when reporting aggregate class-group metrics on the held-out test split. A *case-disjoint split* means that all slides, patches, and features from one TCGA participant stay in exactly one of train, validation, or test. Table 14 in Appendix B lists the same terms in compact form for quick reference.



Field	Example value
Cancer type	Glioblastoma multiforme
Resolution level	0
Patch size	256×256 RGB pixels
Source unit	One crop from a selected tumor region
Patch role	Image-level sample if selected by the controlled manifest
WSI-bag role	Candidate instance in the slide's feature bag

Table 2: Example TCGA-UT data point copied from the shared raw dataset. The image is a single patch, while the table shows how that patch relates to the manifest fields used by the benchmark.

2.2 Shared Splits and Benchmark Inputs

For each seed, the split is assigned at the TCGA participant level rather than at the slide level. Case identifiers are derived from the first three TCGA barcode fields, so all diagnostic slides

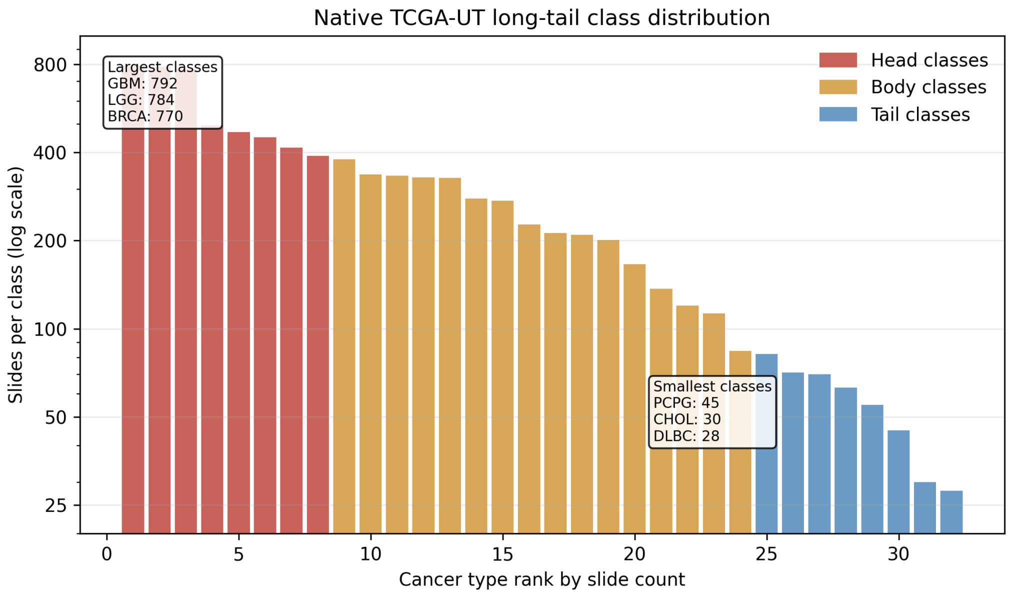


Figure 1: Native TCGA-UT slide support by cancer-type rank. The moderately imbalanced distribution motivates metrics that weight classes more evenly than aggregate accuracy.

from the same participant are kept in the same train, validation, or test partition. Every seed uses the same case-count budget—5,025 training, 1,075 validation, and 1,075 test cases—while the specific participants assigned to each partition differ by seed. Because cases contribute different numbers of diagnostic slides, the resulting slide counts vary across seeds: training contains 6,091–6,129 slides, validation 1,276–1,337 slides, and test 1,299–1,331 slides. The same case-level assignments are reused by both benchmarks, so no patches, frozen features, or WSI bags from the same slide or participant cross train, validation, and test.

This case-disjoint partition is the implemented leakage-control mechanism. It is stricter than slide-disjoint splitting alone, which would prevent patches from the same slide from crossing splits but would still allow different diagnostic slides from the same TCGA participant to appear in different partitions. The generated split reports record `case_id` as the split unit for all three seeds and no cross-split participant leakage after assignment. The patch benchmark and the WSI-bag benchmark see different input units, but they inherit the same train/validation/test case partition.

The patch benchmark derives a deterministic controlled patch manifest from the shared slide split. It uses resolution level 0 and samples up to 30 patches per slide with a fixed rule before method-specific training begins. The cap is a compromise between data coverage and slide-level control: it uses the empirical median number of raw JPG patches per slide at resolution level 0 (median 30, mean 31.1, range 10–80) while preventing slides with many extracted tumor-region patches from dominating the patch-level loss. At this resolution, 8,686 of 8,736 slides have at least 30 raw patches, so the cap retains the typical per-slide patch evidence while limiting only the larger slide folders. Each selected patch is encoded once with a frozen Virchow2 model, and all patch methods train the same lightweight classifier head on these embeddings (Zimmermann et al., 2024; Vorontsov et al., 2024). The encoder output is concatenated as $\mathrm{concat}(\mathrm{CLS}, \mathrm{mean}(\text{patch tokens}))$, yielding 2,560-dimensional float16 vectors that are cast to float32 before classifier training. This design isolates imbalance mechanisms from both patch-selection policy and end-to-end image-backbone learning.

The WSI-bag benchmark applies the same per-slide evidence budget: when a frozen feature bag contains more than 30 instances, it is deterministically subsampled to 30 evenly spaced instances before attention-MIL training and evaluation. This keeps compute bounded and avoids a systematic advantage for slides with larger extracted feature bags. The two benchmarks still differ in supervision unit and aggregation: patch models classify each selected crop independently, whereas WSI-bag models predict one slide label from an attention-weighted set of up to 30 instances.

3 Methods

The methods are organized by the level at which they intervene in the imbalanced learning problem. Cross-entropy (CE) is treated as the no-mitigation reference condition: it uses the observed training distribution as given and does not rebalance samples, labels, costs, representations, or generated data. The remaining methods are tested imbalance interventions grouped according to the taxonomy used in the class-imbalance literature: data-level sampling and augmentation, algorithm-level losses, hybrid sampling–loss methods, synthetic generation, and representation or architecture-level methods (Saini and Susan, 2023; Zhang et al., 2023).

The comparison is regime-specific. Patch methods consume frozen Virchow2 embeddings of controlled patches and optimize one shared MLP classifier head; CE-based patch methods use the same linear output layer, while CFAL replaces that layer with class prototypes and affinity-based prediction. WSI-bag methods consume capped frozen Virchow2 feature bags and optimize one shared attention-MIL head. Both benchmarks therefore rely on the same pathology foundation-model family, but they differ in supervision unit: one embedding per selected patch versus aggregation over a bounded set of instances on a slide. Method differences within a benchmark therefore correspond to imbalance mechanisms rather than changes in split, evaluation code, or encoder choice.

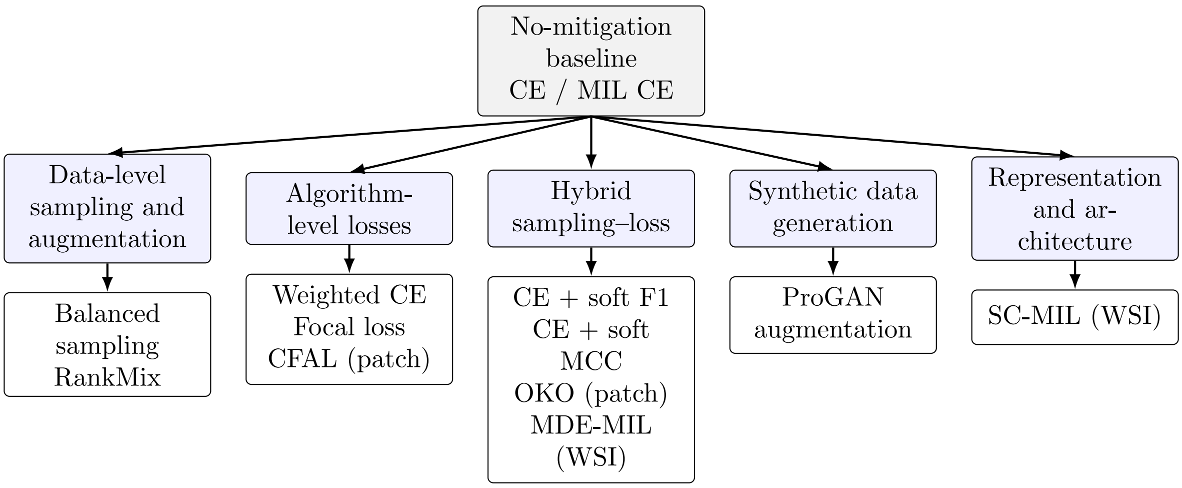


Figure 2: Method organization used in the benchmark. Cross-entropy is the no-mitigation reference, while the tested alternatives intervene through sampling, loss design, synthetic data, WSI feature augmentation, or representation learning.

3.1 Notation

Let C be the number of cancer-type classes and let n_c denote the number of training examples from class c. For a labeled patch or slide-level bag with label $y \in \{1, \ldots, C\}$, the model outputs logits $\mathbf{z} \in \mathbb{R}^C$ and class probabilities

$$p_c = \frac{\exp(z_c)}{\sum_{j=1}^{C} \exp(z_j)}. \tag{1}$$

For WSI-bag methods, a slide is represented as a bag $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ of frozen patch-feature instances. The attention-MIL backbone maps each bag to a bag embedding $\mathbf{b}(B)$ and class logits. Unless a method explicitly changes the sampler, objective, or representation loss, the empirical class frequencies in the training set define the optimization signal.

3.2 Cross-Entropy Baseline

Unweighted cross-entropy is the baseline because it is the standard supervised objective without an explicit class-imbalance correction. For an example with label y, the loss is

$$\mathcal{L}_{\mathrm{CE}}(\mathbf{z}, y) = -\log p_y. \tag{2}$$

In the patch benchmark, this objective trains the shared linear head on frozen Virchow2 patch embeddings from the controlled manifest. In the WSI-bag benchmark, the same objective trains the shared attention-MIL model on slide-level feature bags. In both cases, CE uses the observed class distribution directly: majority classes appear more often in minibatches and therefore contribute more loss terms over an epoch. This makes CE an appropriate reference for asking whether an imbalance-specific intervention actually improves the rare-class tradeoff.

3.3 Data-Level Sampling and Augmentation

Data-level methods change what the optimizer sees, rather than changing the mathematical penalty assigned to one observed example. In the taxonomy, this family includes resampling, augmentation, and WSI-specific feature mixing strategies that make minority or informative examples more visible during training (Saini and Susan, 2023).

3.3.1 Balanced Sampling

Balanced sampling changes minibatch composition by sampling training examples with probabilities inversely related to their class support. For an example i with class y_i, the sampler weight is proportional to

$$q_i \propto \frac{1}{n_{y_i}}. \tag{3}$$

After normalization across the training set, this makes the expected class exposure closer to uniform even though the underlying dataset remains imbalanced. The patch variant applies this sampler to patch examples and still optimizes CE. The MIL variant applies the same idea at slide-bag level and also optimizes CE. Balanced sampling therefore tests a pure data-level intervention: it asks whether equalizing training exposure is enough without changing the loss formula.

3.3.2 RankMix

RankMix is a WSI-native data-augmentation method for weakly supervised slide classification. Ordinary image mixup assumes two inputs with compatible spatial shapes, but WSI bags can contain different numbers of feature instances and lack instance-level labels. RankMix addresses this by mixing representative feature subsequences rather than raw slides (Chen and Lu, 2023).

The tested implementation follows the defining two-stage teacher–student procedure. First, a teacher attention-MIL model is trained with slide labels for 30 epochs using the same MIL architecture and optimization settings as plain MIL CE. For each bag $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ and target class, the teacher assigns ranking scores to instances using its softmax probability for that class; the top-k instances are kept as a representative subsequence, where $k = \min(|B_a|, |B_b|)$ for the two bags being mixed, and their original within-bag order is preserved after selection.

The student MIL model is then reinitialized and trained on bags built from these ranked subsequences.

Mixing itself follows the mixup idea in feature space. For two bags a and b, let H_a and H_b denote the selected instance-feature matrices and $\mathbf{y}_a, \mathbf{y}_b$ the corresponding one-hot slide labels. A scalar mixing coefficient $\lambda \in (0, 1)$ controls how much of each bag is retained:

$$\widetilde{H} = \lambda H_a + (1 - \lambda) H_b, \qquad \widetilde{\mathbf{y}} = \lambda \mathbf{y}_a + (1 - \lambda)\mathbf{y}_b. \tag{4}$$

When λ is close to one, the mixed bag resembles bag a; when it is close to zero, it resembles bag b. Values near $\lambda = \frac{1}{2}$ interpolate more evenly between the two sources. RankMix does not fix λ to a single value. Instead, each mixed example draws

$$\lambda \sim \mathrm{Beta}(\alpha, \alpha), \tag{5}$$

independently. The Beta distribution is a standard choice on the open interval $(0, 1)$ because it is flexible and symmetric when both shape parameters are equal. With density

$$f(\lambda; \alpha, \alpha) \propto \lambda^{\alpha-1}(1 - \lambda)^{\alpha-1}, \qquad \lambda \in (0, 1), \tag{6}$$

the parameter $\alpha > 0$ controls how strongly mixing concentrates around the endpoints versus the center. For $\alpha = 1$, $\mathrm{Beta}(1, 1)$ is the uniform distribution on $(0, 1)$, so every mixing weight is equally likely and the procedure reduces to ordinary symmetric mixup. Larger α pushes λ toward $\frac{1}{2}$, producing more balanced interpolations; smaller α pushes mass toward 0 and 1, producing mixtures that stay closer to one source bag. The reference default is $\alpha = 1.0$, matching the RankMix source implementation; validation selects the best value from Table 5. The student is trained with soft-label cross-entropy on the mixed bags and labels $\widetilde{\mathbf{y}}$. This mechanism is categorized as data-level augmentation because it increases the diversity of the training bags in feature space while retaining the same attention-MIL prediction architecture.

3.4 Algorithm-Level Losses

Algorithm-level methods keep the training examples fixed but alter how errors are penalized. They are cost-sensitive or difficulty-sensitive objectives: minority examples may receive larger explicit weights, or easy examples may contribute less to the gradient (Saini and Susan, 2023; Scholz et al., 2024).

3.4.1 Class-Weighted Cross-Entropy

Class-weighted CE multiplies each example's CE term by a class-dependent cost:

$$\mathcal{L}_{\mathrm{WCE}}(\mathbf{z}, y) = -w_y \log p_y, \qquad w_c \propto \frac{1}{n_c}. \tag{7}$$

The implementation normalizes the inverse-frequency weights so that the average scale of the loss remains comparable across methods. This method leaves the sample order and model architecture unchanged, but increases the loss contribution of classes with fewer training examples. It is evaluated for both patch classification and attention-MIL.

3.4.2 Focal Loss

Focal loss reduces the contribution of confidently classified examples and concentrates optimization on examples that remain difficult. With focusing parameter γ, the multiclass focal objective is

$$\mathcal{L}_{\mathrm{focal}}(\mathbf{z}, y) = -(1 - p_y)^{\gamma} \log p_y. \tag{8}$$

The reference default is $\gamma = 2.0$ for both regimes; validation selects γ from Table 5 while keeping the rest of the training protocol unchanged. The method is relevant to imbalance because many minority-class examples can remain hard under a classifier dominated by head-class gradients, but the focal term does not directly encode the class counts. It is therefore a difficulty-sensitive loss rather than a sampler or explicit prior correction.

3.4.3 Center-Focused Affinity Loss (CFAL)

Center-focused affinity loss (CFAL) replaces the linear softmax head with learnable class prototypes and an affinity-based training objective (Mahbub et al., 2024). The motivation is that minority histology classes can remain visually similar in embedding space even when a frozen encoder already separates them coarsely; pulling examples toward compact class centers and penalizing confusion with nearby wrong-class prototypes can therefore improve tail-class boundaries without changing the sampler.

The tested patch implementation keeps the shared MLP trunk that maps a frozen Virchow2 embedding $\mathbf{x}$ to a hidden representation $\mathbf{z} = f(\mathbf{x}) \in \mathbb{R}^{512}$, but replaces the final linear map with one learnable prototype $\mathbf{w}_c \in \mathbb{R}^{512}$ per class. Both embeddings and prototypes are L2-normalized before affinity computation. The Gaussian affinity between example i and class c is

$$a_{ic} = \exp\left(-\frac{\|\hat{\mathbf{z}}_i - \hat{\mathbf{w}}_c\|_2^2}{\sigma}\right), \qquad \hat{\mathbf{v}} = \frac{\mathbf{v}}{\|\mathbf{v}\|_2}, \tag{9}$$

where σ controls how quickly affinity decays with prototype distance. At inference, class scores are $\log a_{ic}$ followed by softmax normalization. The affinities $a_{ic} \in (0, 1]$ are geometric similarity scores, not class probabilities: they are trained with a margin objective and never optimized to satisfy likelihood or calibration constraints. Applying softmax to $\log a_{ic}$ therefore yields a convenient ranking score for argmax decisions, but it should not be read as a calibrated estimate of $P(y = c \mid \mathbf{x})$ without additional post-hoc mapping.

Training optimizes a margin term on affinities rather than cross-entropy on logits. For batch index i with label y_i, define the per-example margin loss

$$\ell_i = \sum_{c \neq y_i} \max\big(0,\, \lambda + a_{ic} - a_{i,y_i}\big), \tag{10}$$

which penalizes wrong-class affinities that exceed the true-class affinity by less than margin λ. A diversity regularizer $R(\mathbf{W})$ encourages prototypes to spread apart by penalizing deviations of pairwise squared distances from their mean. The total objective combines class reweighting, a center-focused local weight, and prototype diversity:

$$\mathcal{L}_{\text{CFAL}} = \frac{1}{B} \sum_{i=1}^{B} \frac{1}{E_{y_i}} (1 - a_{i,y_i})^{\gamma}\, \ell_i + R(\mathbf{W}), \tag{11}$$

where E_c is the effective number of training examples in class c derived from class count n_c and hyperparameter β, and γ up-weights examples whose true-class affinity remains low. Unlike the CE plus soft-F1 and soft-MCC hybrids, CFAL does not use balanced oversampling: it is categorized as an algorithm-level patch method because it changes the head and loss while keeping the natural training distribution.

The reference defaults are $\lambda = 0.1$, $\beta = 0.999$, and $\gamma = 2.0$ following Mahbub et al.; all three are held fixed and validation instead selects the affinity bandwidth σ from Table 5. The paper's $\sigma = 130$ was calibrated for un-normalized AlexNet FC features; on L2-normalized 2,560-dimensional Virchow2 vectors that scale is far too large and collapses all pairwise affinities, so this benchmark evaluates a range matched to the normalized embedding geometry. CFAL is evaluated only in the patch regime because the source method assumes instance-level labels rather than weak slide-level bags.

3.5 Hybrid Sampling–Loss Methods

Hybrid methods combine data-level and algorithm-level changes. The Scholz-style patch objectives below adapt the metric-derived losses of Scholz et al. for imbalanced medical classification (Scholz et al., 2024). Both tested variants use class-balanced oversampling together with metric-derived losses, so their effect cannot be attributed only to a new loss or only to a new sampler.

3.5.1 Scholz-Style CE + Soft F1

The intuition is that CE optimizes the log-probability of the correct class one example at a time, whereas macro F1 evaluates a different object: the balance between precision and recall at class level. Under class imbalance, this distinction matters because a model can reduce CE substantially by fitting frequent classes while still leaving tail-class recall or precision weak.

The obstacle is that ordinary F1 is computed from hard counts after prediction, so it is not directly usable as a gradient-based training loss. The Scholz-style soft-F1 objective replaces those hard counts with probability-weighted counts inside each minibatch. For one class, a confident correct prediction contributes almost one soft true positive, a confident prediction for the wrong class contributes to soft false positives, and low probability on a true class contributes to soft false negatives. Formally, for one-hot labels y_i and predicted probabilities $\hat{y}_i$:

$$\mathrm{TP} = \sum_i y_i \hat{y}_i, \quad \mathrm{FP} = \sum_i (1 - y_i)\hat{y}_i, \quad \mathrm{FN} = \sum_i y_i (1 - \hat{y}_i). \tag{12}$$

These terms keep the familiar meaning of a confusion matrix but make every component differentiable with respect to the model probabilities. They then define soft precision, soft recall, and soft F1:

$$\mathrm{F1}_{\mathrm{soft}} = \frac{2\,\mathrm{precision}_{\mathrm{soft}}\,\mathrm{recall}_{\mathrm{soft}}}{\mathrm{precision}_{\mathrm{soft}} + \mathrm{recall}_{\mathrm{soft}}}. \tag{13}$$

The multiclass implementation applies this construction separately to each class and averages the resulting classwise terms. This macro-style averaging is the imbalance-aware part: each class contributes one term regardless of how many samples it has in the full dataset. The tested objective does not replace CE completely; it adds the metric-derived term to CE so that optimization still rewards calibrated class probabilities while also pushing the minibatch toward higher macro soft F1:

$$\mathcal{L}_{\mathrm{CE+F1}} = \mathcal{L}_{\mathrm{CE}} + \left(1 - \frac{1}{C}\sum_{c=1}^{C} \mathrm{F1}_{\mathrm{soft},c}\right). \tag{14}$$

The benchmark trains this loss with balanced sampling on the patch manifest. That design is important for interpretation: the loss tries to optimize a macro-style class metric, while the sampler ensures that tail classes occur often enough in minibatches for their soft-F1 terms to provide a regular training signal.

3.5.2 Scholz-Style CE + Soft MCC

The soft-MCC variant uses the same principle as soft F1—replace hard evaluation counts with differentiable probability masses—but starts from the Matthews correlation coefficient (MCC). Whereas F1 for one class depends mainly on that class's positive predictions, MCC is built from the full confusion matrix and therefore reacts to false positives, false negatives, and true negatives at the same time. In binary problems it is often preferred in imbalanced settings because it can remain informative when one class dominates the dataset: accuracy can look high while predictions are still systematically biased, but MCC penalizes both kinds of mistakes jointly.

Binary MCC as a correlation coefficient. For one class treated as positive, write TP, TN, FP, and FN for the four confusion-matrix cells. The standard MCC is

$$\mathrm{MCC} = \frac{\mathrm{TP}\cdot\mathrm{TN} - \mathrm{FP}\cdot\mathrm{FN}}{\sqrt{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}}. \tag{15}$$

The numerator compares co-occurrence of correct positive and correct negative decisions with co-occurrence of the two error types; the denominator scales by how many examples fall into each margin of the confusion matrix. MCC equals +1 for perfect predictions, 0 for chance-level association, and can become negative when predictions are worse than random. This makes it a correlation-style score rather than a prevalence-weighted rate.

Multiclass MCC in a minibatch. For C cancer types, the multiclass extension treats labels and predictions as C-dimensional vectors per example. Let $Y \in \{0,1\}^{N\times C}$ be the one-hot label matrix and $\widehat{Y} \in [0,1]^{N\times C}$ the corresponding softmax probabilities, with rows $\mathbf{y}_i$ and $\hat{\mathbf{y}}_i$. Three aggregate quantities describe how mass is distributed inside the batch:

$$s = \sum_{i=1}^{N}\sum_{c=1}^{C} Y_{ic}\widehat{Y}_{ic}, \qquad t_c = \sum_{i=1}^{N} Y_{ic}, \qquad p_c = \sum_{i=1}^{N} \widehat{Y}_{ic}. \tag{16}$$

Here s is the total probability-weighted agreement between labels and predictions: each example contributes $\hat{y}_{i,y_i}$, the predicted mass on its true class. The terms t_c and p_c are the batch totals of true and predicted mass for class c. They play the role of marginals in the confusion-matrix calculation. If predictions collapse onto a few head classes, the vector $(p_1,\ldots,p_C)$ becomes skewed even when argmax accuracy looks acceptable; MCC is sensitive to that mismatch because it compares the joint alignment s with what would be expected from the marginals alone.

The multiclass MCC used in training can be written as a normalized covariance between the label matrix and the prediction matrix:

$$\mathrm{MCC}_{\mathrm{mc}} = \frac{Ns - \sum_{c=1}^{C} t_c p_c}{\sqrt{N^2 - \sum_{c=1}^{C} p_c^2}\,\sqrt{N^2 - \sum_{c=1}^{C} t_c^2}}. \tag{17}$$

The numerator $Ns - \sum_c t_c p_c$ increases when true and predicted mass line up example by example and decreases when correct predictions are no better than what the marginal totals would suggest by chance. The two square-root terms measure how spread out the batch-level label totals and prediction totals are across classes; they prevent the score from exploding when one minibatch happens to contain only a few classes. Equation (17) reduces to the familiar binary form in (15) when $C = 2$.

Soft MCC loss. Hard MCC replaces $\widehat{Y}$ by one-hot argmax predictions, which again blocks gradient-based training. The soft-MCC objective keeps the structure of (17) but plugs in softmax probabilities directly:

$$\mathrm{MCC}_{\mathrm{soft}} = \frac{Ns - \sum_{c=1}^{C} t_c p_c}{\sqrt{N^2 - \sum_{c=1}^{C} p_c^2}\,\sqrt{N^2 - \sum_{c=1}^{C} t_c^2}}. \tag{18}$$

Every term in (16) and (18) is differentiable with respect to the logits that produce $\widehat{Y}$. A confident correct prediction increases s; probability placed on the wrong classes increases some p_c while leaving the matching t_c unchanged, which lowers the numerator; and overly peaked prediction vectors reduce the prediction-side denominator term, limiting the score even if a few examples are classified correctly. As with soft F1, the training objective turns a score to be maximized into a loss to be minimized and adds it to CE:

$$\mathcal{L}_{\text{CE+MCC}} = \mathcal{L}_{\text{CE}} + (1 - \text{MCC}_{\text{soft}})\,. \tag{19}$$

This variant is also evaluated only in the patch benchmark and is trained with class-balanced oversampling. The soft-F1 and soft-MCC variants therefore ask a focused question: once tail classes are regularly presented to the optimizer, does adding a differentiable version of a class-balanced evaluation metric improve the controlled patch classifier beyond CE and simple balanced sampling?

3.5.3 Odd-K-Out (OKO) Set Learning

Standard cross-entropy optimizes each example independently, which has two relevant consequences under class imbalance. First, the gradient signal for a tail class arrives only when that class is sampled, so rare examples can be overwhelmed by frequent ones. Second, hard-label cross-entropy encourages logit values to diverge without bound, which is a known driver of overconfidence and poor calibration (Guo et al., 2017). OKO (Muttenthaler et al., 2024) addresses both problems simultaneously by replacing per-example cross-entropy with a set-level objective. Calibration is inherently a property of a model's output distribution over a collection of examples, and training on sets is a natural way to expose the model to that distributional structure during learning.

Set construction. A training set S of size $k+2$ is constructed as follows. A pair class y' is sampled uniformly from $[C]$, equalizing class frequency in expectation regardless of the training distribution—analogous to balanced sampling but applied at the set level rather than the sample level. Two examples x'_1, x'_2 are drawn from the class y', and k distinct odd classes $y'_3, \ldots, y'_{k+2}$ are sampled without replacement from $[C] \setminus \{y'\}$. One representative example x'_j is drawn from each odd class y'_j. The resulting set $S = (x'_1, \ldots, x'_{k+2})$ therefore always contains exactly one matched pair and k distractors from different classes.

Set-aggregated loss. For a model f_θ parameterized by θ, OKO defines the aggregated logit vector as the sum of individual per-example outputs over the set members:

$$f_\theta(S) = \sum_{i=1}^{k+2} f_\theta(x'_i). \tag{20}$$

Prediction on the aggregated logits uses ordinary softmax. The pair class y' appears twice in the set and its two logit contributions therefore reinforce each other, raising the aggregate score for y' relative to any single odd class. This pooling effectively averages the raw logit values across set members, which limits how extreme the aggregate can be—an implicit logit regularization that the paper links to improved calibration even when training with hard labels.

The main training loss predicts the pair class from the aggregated logits:

$$\ell_{\text{OKO}} = -\log \text{softmax}(f_\theta(S))_{y'}. \tag{21}$$

Following the main-text configuration of the original paper, the implementation also trains an auxiliary classification head g_θ that shares the trunk with f_θ and predicts the first odd class y'_3:

$$\ell_{\text{aux}} = -\log \text{softmax}(g_\theta(S))_{y'_3}. \tag{22}$$

The total per-set loss is $\ell_{\text{OKO}} + \ell_{\text{aux}}$. The auxiliary head is discarded at inference, and single-example evaluation proceeds exactly as for any other classifier: the model accepts one patch feature vector and returns the logits from the main head. There is no inference overhead or architecture change at test time.

OKO belongs in the hybrid sampling–loss category because the set construction couples a balanced class-sampling policy with a modified training objective; neither component is meaningful on its own. Balanced sampling via uniform pair-class selection ensures that all C classes contribute equally to the gradient signal in expectation, which directly addresses the data-level imbalance problem. The aggregated set-level loss then introduces an implicit regularization on logit magnitude. Together these are not separable in the same way that standard cross-entropy and a WeightedRandomSampler are separable. In this benchmark OKO is evaluated in the patch feature regime only; the set-construction mechanism does not extend naturally to the attention-MIL bag encoder used in the WSI-bag benchmark.

The number of odd classes k controls set size and therefore the degree of logit averaging. The paper's ablation experiments show that accuracy and calibration are insensitive to the specific value of k, and the default of $k = 1$ is the computationally cheapest option. The validation sweep covers $k \in \{1, 2, 3, 4, 5, 6, 8, 12, 20\}$ with the default fixed at $k = 1$. The validation-selected configuration reported later is $k = 4$ (Table 6), indicating that a moderate set size works best in this benchmark without requiring the largest sets.

3.6 Synthetic Data Generation

Synthetic generation is data-level in spirit but is separated because it introduces new generated samples rather than resampling observed ones. In histopathology, this family is attractive because deleting majority-class tissue wastes data, while naive image-space interpolation often produces unrealistic pathology images (Saini and Susan, 2023; Ruiz-Casado et al., 2024). The practical hope is simple: if real minority-class patches are scarce, a generator may learn enough of their stain, texture, and tissue morphology distribution to provide additional plausible training examples.

3.6.1 ProGAN Augmentation

The ProGAN experiment adapts the GAN-based histopathology augmentation idea of Ruiz-Casado et al. to multiclass TCGA-UT patches (Ruiz-Casado et al., 2024). A generative adversarial network (GAN) has two coupled models. The generator G maps random latent vectors to synthetic images, while the discriminator D tries to distinguish real training patches from generated patches. Training alternates between making D better at this distinction and making G produce images that D cannot reliably reject. The standard objective expresses this competition as

$$\min_G \max_D \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{u}\sim p(\mathbf{u})}[\log(1 - D(G(\mathbf{u})))]. \tag{23}$$

This equation is not a classification loss for the final TCGA-UT classifier. It is the pretraining objective for producing additional patch images. After generation, the classifier simply sees a larger patch training set.

ProGAN modifies ordinary GAN training by growing the image resolution gradually. Instead of asking the generator to synthesize 256×256 histology patches from the start, training begins at a coarse resolution where global color and texture structure is easier to model. Higher-resolution layers are then introduced step by step until the target patch size is reached. This staged procedure is intended to stabilize synthesis and reduce the risk that the generator learns only a narrow set of repeated images. The implementation uses the Wasserstein gradient penalty (WGAN-GP, $\lambda = 10$, drift $= 10^{-3}$) in place of the non-saturating binary cross-entropy objective in (23), and applies equalized learning rate (weight $\sim \mathcal{N}(0, 1)$; scale $\sqrt{2/\text{fan\_in}}$ applied per layer at forward time), matching the `pro_gan_pytorch` configuration used by Ruiz-Casado et al. Progressive growth, PixelNorm, and minibatch-standard-deviation are retained. In this benchmark, one class-specific generator is trained for every training class whose patch count is below the head-class training count. Table 3 summarizes the generator training settings adapted from Ruiz-Casado et al. to multiclass TCGA-UT patch augmentation.

Setting	Value
Resolution stages	4×4 to 256×256 (seven stages)
Epochs per stage (depths 1–6)	25
Final-depth (256×256) refinement epochs	25 (reference); validation selects from $\{10, 25, 50\}$
Fade-in fraction for new layers	0.5 (pinned to the reference 25-epoch schedule)
Latent dimension	128
Base channel width	512
Adversarial loss	WGAN-GP ($\lambda = 10$, drift $= 10^{-3}$)
Equalized learning rate	Yes (weight $\sim \mathcal{N}(0, 1)$, scale $\sqrt{2/\text{fan\_in}}$ at forward)
Optimizer	Adam, learning rate 2×10^{-4}, $\beta_1 = 0.5$
Batch schedule	Depth-dependent schedule of Ruiz-Casado et al.
Real patches per class for GAN training	Up to 2,048, deterministic per-class subsample
Synthetic count per tail class	Head-class train count minus class train count

Table 3: ProGAN generator training settings. Depths 1–6 use 25 epochs each. For the final depth, one long run trains to 50 epochs and the generator is snapshotted at epochs 10, 25, and 50; validation selects the snapshot with the best mean macro F1 over seeds. This reuses the single GAN training run for all three sweep points at roughly 2× the previous final-depth cost. Synthetic fraction is fixed at 1 (every tail class is balanced to the head-class count).

When a class has more than 2,048 real training patches, at most 2,048 are drawn uniformly at random with a deterministic per-class seed for ProGAN training only; the downstream classifier still uses the full real training set together with the generated patches. Inception FID is computed for every augmented class between the generated images and the real training subset used for that class-specific generator, but no generated samples are filtered by FID or other quality thresholds. Real and synthetic RGB patches are embedded with the same frozen Virchow2 encoder before classifier training. The synthetic fraction is fixed at 1 so each tail class is balanced to the training-set head count; validation instead selects the number of final-depth (256×256) refinement epochs from the grid in Table 5. Synthetic images are used only as extra patch-classification training examples; they are not converted into synthetic WSI bags.

3.7 Representation and Architecture-Level Methods

Representation-level methods try to make rare classes more separable in feature space rather than only changing sampling or per-example loss weights. This family is especially relevant for computational pathology, where WSI labels supervise bags of many unlabeled instances and class imbalance can occur both across slides and within a positive slide (Juyal et al., 2024).

3.7.1 SC-MIL

SC-MIL extends supervised contrastive learning to MIL by applying the contrastive objective to bag representations instead of assigning noisy slide labels to individual patches (Juyal et al., 2024). The motivation is that a slide label usually does not describe every patch in a bag. A cancer-type label can be correct for the slide while many instances contain background, normal tissue, technical noise, or weakly informative morphology. Applying contrastive learning directly to instances would therefore treat many patches as positives or negatives using labels they do not truly carry. SC-MIL avoids that mismatch by contrasting the aggregated slide-level representation.

The attention-MIL backbone first maps bag B_i to an embedding $\mathbf{b}_i = \mathbf{b}(B_i)$. A projection head g then produces normalized contrastive features $\mathbf{r}_i = g(\mathbf{b}_i)$. In the contrastive view, each bag in a minibatch is an anchor. Bags with the same cancer-type label are positives because their slide-level representations should become closer, while bags from other classes act as negatives because their representations should remain separated. For anchor i, bags of the same class form the positive set $P(i)$, and the supervised contrastive term is

$$\mathcal{L}_{\text{SCL}} = -\frac{1}{\sum_i |P(i)|} \sum_i \sum_{p \in P(i)} \log \frac{\exp(\mathbf{r}_i^\top \mathbf{r}_p/\tau)}{\sum_{a \neq i} \exp(\mathbf{r}_i^\top \mathbf{r}_a/\tau)}. \tag{24}$$

The numerator rewards similarity to same-class bags. The denominator compares that similarity against all other bags in the minibatch, so the loss favors representations where class clusters are compact and mutually separated. The temperature τ controls how sharply these similarities are weighted; the reference default is $\tau = 1.0$, and validation selects τ from Table 5.

SC-MIL does not rely on contrastive learning alone. The contrastive term shapes the representation space, while CE trains the final class predictor. The tested variant combines them through a linear curriculum:

$$\mathcal{L}_t = \beta_t \mathcal{L}_{\text{SCL}} + (1 - \beta_t)\mathcal{L}_{\text{CE}}, \qquad \beta_t = 1 - \text{progress}_t. \tag{25}$$

Early training therefore emphasizes class-structured bag embeddings, when the classifier head is still unstable. As training progresses, β_t decreases and more weight moves to CE, so the learned representation is tied back to the actual slide-level classification objective. Activation diagnostics count positive pairs during training to confirm that the contrastive component is active.

3.7.2 MDE-MIL

MDE-MIL addresses long-tailed WSI classification with distribution-specific experts and a cross-expert consistency term (Ling et al., 2025). The full source method combines a dual-expert ensemble with multimodal distillation from a pathology foundation model; this benchmark implements the ensemble branch only—shared attention aggregator, two expert classification heads, and a logit consistency loss—under the same frozen Virchow2 bags as the other MIL methods, without CONCH multimodal distillation.

The shared trunk matches plain attention MIL: each bag $B = \{\mathbf{x}_1, \ldots, \mathbf{x}_m\}$ is encoded instance-wise, attention weights are computed with a linear scorer and softmax over instances, and the weighted sum yields a bag embedding $S(B) \in \mathbb{R}^d$. Two separate feed-forward experts then map S to logits:

$$E_U(S) = W_U\, \phi_U(S) + b_U, \qquad E_B(S) = W_B\, \phi_B(S) + b_B, \tag{26}$$

where ϕ_U and ϕ_B are two-layer MLPs with ReLU and dropout. Expert U is trained on the natural slide distribution and expert B on a class-balanced slide distribution.

Each training step draws one minibatch from each distribution. The unbalanced loader shuffles slide bags as in MIL CE; the balanced loader uses the same inverse-frequency weights as balanced-sampling MIL. For batches (B_U, y_U) and (B_B, y_B), the classification loss is

$$\mathcal{L}_{\text{cls}} = \text{CE}\big(E_U(S(B_U)), y_U\big) + \text{CE}\big(E_B(S(B_B)), y_B\big), \tag{27}$$

and the cross-expert consistency term penalizes disagreement on the *same* bag embedding:

$$\mathcal{L}_{\text{con}} = \text{MSE}\big(E_U(S(B_U)), E_B(S(B_U))\big) + \text{MSE}\big(E_B(S(B_B)), E_U(S(B_B))\big). \tag{28}$$

The total objective is $\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_{\text{con}}\mathcal{L}_{\text{con}}$. The reference default is $\lambda_{\text{con}} = 0.25$, matching the Camelyon+-LT default reported by Ling et al.; validation selects λ_{con} from Table 5. Unlike

RankMix, MDE-MIL does not use a teacher model or student reinitialization: both experts share the aggregator throughout training.

At validation and test time, each slide bag is evaluated once in natural order with mean-logit ensemble inference:

$$\hat{\mathbf{z}}(B) = \tfrac{1}{2}\big(E_U(S(B)) + E_B(S(B))\big). \tag{29}$$

MDE-MIL is categorized as a hybrid WSI method because it couples two sampling distributions with a consistency regularizer rather than changing only the sampler or only the per-example loss.

3.8 Training Protocol and Reproducibility

Each method is evaluated under one shared training budget per benchmark (Table 4). Cross-entropy baselines use cited defaults without an imbalance-specific sweep. All other methods vary one imbalance-specific control on validation; the configuration with the highest mean validation macro F1 over seeds 0–2 is selected, with balanced accuracy as a tie-breaker. The test split never informs selection. Table 5 lists the reference default and validation grid for each control; the selected configurations and their tuning sensitivity are reported with the results (Section 5.1). ProGAN sweeps the number of final-depth (256×256) refinement epochs (Table 3); one long run produces snapshots at epochs 10, 25, and 50 so all sweep points share a single GAN training cost. The synthetic fraction is fixed at 1 (balance each tail class to the head-class count).

Setting	Patch benchmark	WSI-bag benchmark
Optimizer	AdamW	AdamW
Learning rate	10^{-3}	10^{-3}
Weight decay	10^{-4}	10^{-4}
Epochs	30	30
Early stopping	none	none
Minibatch size	256 patches	32 slide bags
Model selection	final epoch	final epoch
Random seeds	0, 1, 2	0, 1, 2

Table 4: Shared training protocol for both benchmarks. All methods within a benchmark use the same optimizer, schedule, and epoch budget unless noted in a method section.

Regime	Method	Reference default	Validation grid
Patch	Weighted CE	Weight power = 1	$\{0.25, 0.5, 0.75, 1\}$
WSI bag	Weighted CE	Weight power = 1	$\{0, 0.125, 0.25, 0.5, 0.75, 1\}$
Patch, WSI	Balanced sampler	Sampler power = 1	$\{0.5, 0.75, 1\}$
Patch, WSI	Focal loss	$\gamma = 2$	$\gamma \in \{0.5, 1, 1.5, 2\}$
Patch	CE + soft F1	Metric-loss weight = 1	$\{0.25, 0.5, 1, 2, 4, 8, 16, 32, 64, 256, 512, 1024\}$
Patch	CE + soft MCC	Metric-loss weight = 1	$\{0.25, 0.5, 1, 2, 8, 32\}$
Patch	CFAL	$\gamma = 2$, $\sigma = 1$	$\sigma \in \{0.25, 1, 4\}$
Patch	ProGAN augmentation	Final-depth epochs = 25	$\{10, 25, 50\}$
Patch	OKO	$k = 1$	$k \in \{1, 2, 3, 4, 5, 6, 8, 12, 20\}$
WSI bag	RankMix	$\alpha = 1$	$\alpha \in \{0.2, 0.5, 1, 2, 4, 8, 16, 32, 128\}$
WSI bag	SC-MIL	$\tau = 1$	$\tau \in \{0.05, 0.1, 0.2, 0.5\}$
WSI bag	MDE-MIL	$\lambda_{\mathrm{con}} = 0.25$	$\lambda_{\mathrm{con}} \in \{0.1, 0.25, 0.3, 0.5, 1, 2, 4\}$

Table 5: Method-specific imbalance controls. Each row varies one control on validation while retaining the shared budget in Table 4. ProGAN sweeps only the final-depth refinement epochs; generator training follows Table 3.

Figure 3 shows the patch and WSI-bag classifier heads.

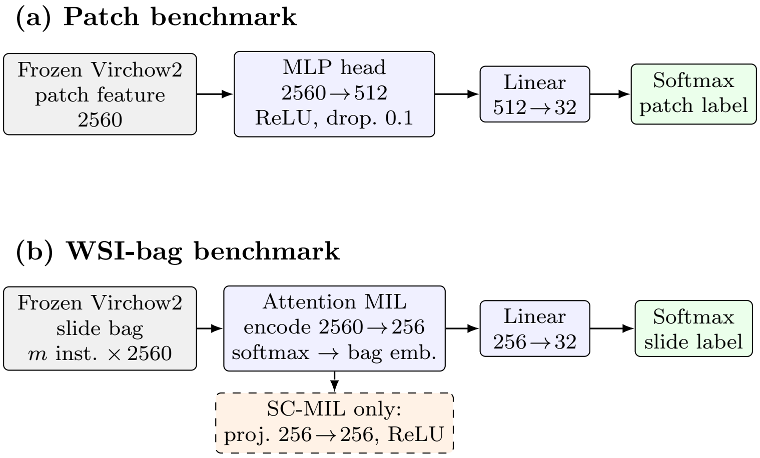


Figure 3: Shared classifier heads for the benchmark. Panel (a) classifies one controlled patch embedding per forward pass. CFAL reuses the MLP trunk but replaces the linear $512 \to 32$ head with learnable prototypes and affinity-based prediction. Panel (b) encodes all instances on a slide, pools them with attention, and predicts one slide-level label; the dashed branch is used only by SC-MIL. MDE-MIL reuses the same attention trunk but replaces the single linear head with two expert heads and mean-logit ensemble inference at evaluation. Virchow2 features are frozen in both benchmarks.

The patch head is a two-layer MLP: linear map to 512 units, ReLU, dropout rate 0.1, and a linear output to 32 classes for CE-based methods. CFAL keeps the same trunk but replaces the final linear map with one learnable prototype per class and Gaussian affinities as in (9). The WSI-bag head is an attention-based MIL network on the same feature dimension: each instance passes through the shared encoder, instance weights come from a linear scorer followed by a softmax over instances, the bag representation is their weighted sum, and slide logits are produced by a final linear map. SC-MIL additionally routes bag embeddings through a two-layer projection head of width 256 with ReLU. MDE-MIL keeps the shared encoder and attention pooling but uses two separate expert heads on the bag embedding. The WSI-bag input is capped at 30 instances per slide using deterministic evenly spaced subsampling.

RankMix uses balanced slide-bag sampling during student training. Mixed bags are formed within each minibatch by pairing bags with a random permutation and mixing their teacher-ranked representative subsequences with $\lambda \sim \mathrm{Beta}(\alpha, \alpha)$ as in (5). The student is reinitialized after teacher preparation and trained for the same 30-epoch budget as the other MIL methods. MDE-MIL instead zips one natural-order batch with one balanced batch at every step, trains both experts on the shared aggregator without a teacher, and evaluates with mean-logit ensembling as in (29).

Head, body, and tail metrics group classes by support in the full TCGA-UT slide distribution rather than by held-out test counts. Cancer types are sorted by dataset slide count; the eight lowest-support classes form the tail tier, the eight highest-support classes form the head tier, and the remaining sixteen classes form the body tier. Tier-wise precision, recall, and F1 on validation and test splits then aggregate only over the predefined classes in each tier.

4 Evaluation

Both benchmarks use seeds 0, 1, and 2. Macro F1 is the primary ranking metric within each benchmark because it weights classes equally and penalizes failure on rare classes. Balanced accuracy, accuracy, classwise precision/recall/F1, and head/body/tail metrics are reported in the main result tables using the tier definitions in Section 3.8. These discrimination metrics answer whether the predicted class is usually correct, but they do not by themselves say whether the reported class probabilities are trustworthy. A model can rank well on macro F1 while assigning near-certainty to many wrong predictions, which is especially relevant in imbalanced medical classification where deployment and thresholding may depend on confidence rather than only on the argmax label (Mosquera et al., 2024). The study therefore reports three complementary probability-quality metrics on held-out softmax outputs; their definitions are given below and their values are reported both before and after the unified post-hoc temperature-scaling protocol. Patch and WSI-bag methods are never ranked against each other, because they answer related but different questions with different input units and supervision.

All reported results use validation-selected configurations evaluated on the held-out test split and show mean ± standard deviation over seeds. Tables 7 and 9 are the main comparison tables; Table 6 records the selected controls. Table 12 reports paired seed-wise differences for headline comparisons against each regime's CE baseline; bracketed values list deltas in seed order 0–2 without formal significance testing or bootstrap confidence intervals. Cross-method gaps should therefore be interpreted from the held-out means together with the paired seed-wise directions. Large gaps with the same paired direction in all seeds provide stronger evidence; WSI-bag differences with overlapping standard deviations remain descriptive even when one method has the highest mean.

4.1 Probability Quality and Calibration

Let N denote the number of held-out examples in one benchmark split, $C = 32$ the number of cancer types, $y_i \in \{1, \dots, C\}$ the true label, and $\hat{\mathbf{p}}_i \in [0,1]^C$ the corresponding softmax probability vector with $\sum_c \hat{p}_{ic} = 1$. Write $Y_i \in \{0,1\}^C$ for the one-hot encoding of y_i. The same three metrics are computed once from raw final-epoch probabilities and once after the required post-hoc temperature-scaling protocol.

Negative log likelihood. Cross-entropy training optimizes the log-probability of the correct class. A natural held-out analogue is the average negative log likelihood (NLL), also called multiclass log loss:

$$\mathrm{NLL} = -\frac{1}{N}\sum_{i=1}^{N} \log \hat{p}_{i,y_i}. \tag{30}$$

Lower NLL means the model assigns higher probability mass to the true class on average. NLL therefore measures both whether the top prediction is correct and how sharply probability is placed on that class: a correct but overconfident model and a correct but cautious model can have different NLL even when their hard-error rates match. In implementation, $\hat{p}_{i,y_i}$ is clipped to $[10^{-12}, 1]$ before taking the logarithm so that numerical underflow does not dominate the score when a class receives extremely small probability.

Multiclass Brier score. The Brier score was introduced for probabilistic forecasts and penalizes the entire predicted distribution rather than only the argmax label (Brier, 1950). For multiclass problems it is the mean squared error between $\hat{\mathbf{p}}_i$ and Y_i:

$$\mathrm{BS} = \frac{1}{N}\sum_{i=1}^{N}\sum_{c=1}^{C}\big(\hat{p}_{ic} - \mathbb{1}[y_i = c]\big)^2 = \frac{1}{N}\sum_{i=1}^{N} \|\hat{\mathbf{p}}_i - Y_i\|_2^2. \tag{31}$$

Lower BS is better. The score is minimized when $\hat{\mathbf{p}}_i$ equals the true class indicator and increases when probability leaks into incorrect classes. Unlike NLL, BS is bounded and symmetric in probability errors across classes, so it is less dominated by a single very small $\hat{p}_{i,y_i}$. Together, NLL and BS evaluate overall distributional fit on the test split.

Expected calibration error. Discrimination and distributional fit can still leave confidence misaligned with empirical accuracy. Expected calibration error (ECE) measures that gap using confidence bins (Naeini et al., 2015; Guo et al., 2017). For each example define the predicted class $\hat{y}_i = \arg\max_c \hat{p}_{ic}$ and the confidence $c_i = \max_c \hat{p}_{ic}$. Partition examples into $B = 10$ equal-width bins on $[0, 1]$ with edges $\tau_0 = 0 < \tau_1 < \cdots < \tau_{10} = 1$ and intervals $(\tau_{b-1}, \tau_b]$. Let $B_b = \{i : c_i \in (\tau_{b-1}, \tau_b]\}$. The bin accuracy and mean confidence are

$$\mathrm{acc}(B_b) = \frac{1}{|B_b|} \sum_{i \in B_b} \mathbb{1}[\hat{y}_i = y_i], \qquad \bar{c}(B_b) = \frac{1}{|B_b|} \sum_{i \in B_b} c_i. \tag{32}$$

ECE is the size-weighted average absolute gap between these quantities:

$$\mathrm{ECE} = \sum_{b=1}^{B} \frac{|B_b|}{N} \left| \mathrm{acc}(B_b) - \bar{c}(B_b) \right|. \tag{33}$$

Lower ECE is better. Intuitively, if a model reports $c_i \approx 0.9$ on many examples, then roughly 90% of those examples should be correct; large systematic deviations indicate over- or under-confidence. ECE is coarse—it depends on binning and uses only the maximum predicted probability—but it is useful here because it separates confidence alignment from headline discrimination and highlights cases where a method improves macro F1 without improving the reliability of its reported probabilities. Reliability diagrams plot empirical accuracy against mean predicted confidence within each bin and make systematic over- or under-confidence visible as departures from the diagonal.

Post-hoc temperature calibration. Calibration is evaluated with one unified post-hoc protocol for every method. For each method and seed, a single positive temperature T is fit on the validation split by minimizing NLL, then applied unchanged to that seed's held-out test score matrix before softmax (Guo et al., 2017). The temperature-scaled table is the decision-facing probability comparison; the raw final-epoch probabilities are retained in Table 13 only as a diagnostic of the uncalibrated model outputs. Temperature scaling leaves argmax rankings unchanged but corrects confidence scale, which is especially important for CFAL because its affinity outputs are ranking scores rather than likelihood-trained probabilities.

5 Results

Both benchmarks share case-disjoint splits, the Virchow2 feature family, and a maximum of 30 feature instances per slide, but they answer different prediction questions. The patch benchmark treats each selected crop as an independent labeled example; the WSI-bag benchmark predicts one slide label from an attention-weighted aggregation over the selected instances on that slide. Patch metrics therefore characterize patch-level classification only and must not be read as slide-level diagnostics; WSI-bag metrics characterize slide-level MIL only. Absolute scores and method orderings are not comparable across the two tables. Supporting diagnostic tables and figures referenced from this section are collected in Appendix A.

5.1 Selected Controls and Tuning Sensitivity

Each non-baseline method contributes one validation-selected imbalance control to the comparison. Table 6 records the control selected for each method, chosen by mean validation macro F1 over seeds 0–2 with balanced accuracy as a tie-breaker (Section 3.8); these configurations define the held-out test results reported in the remaining tables.

How much the selected control matters varies sharply across methods. A seed-blocked repeated-measures analysis of the validation sweeps shows that across-configuration variation exceeds

Regime	Method	Selected control	Val macro F1	Test macro F1	Test bal. acc.
Patch	CE	baseline (no sweep)	–	0.773	0.764
Patch	Balanced sampler	sampler_power=1	0.768	0.771	0.769
Patch	CE + soft F1 (balanced)	metric_loss_weight=256	0.781	0.780	0.781
Patch	CE + soft MCC (balanced)	metric_loss_weight=32	0.771	0.770	0.772
Patch	CFAL	cfal_sigma=1	0.792	0.798	0.789
Patch	Focal	focal_gamma=1.5	0.764	0.765	0.758
Patch	OKO	oko_k=4	0.783	0.788	0.791
Patch	ProGAN augmentation	final_depth_epochs=10	0.756	0.766	0.760
Patch	Weighted CE	weight_power=0.5	0.764	0.774	0.767
WSI bag	MIL CE	baseline (no sweep)	–	0.826	0.823
WSI bag	MDE-MIL (ensemble)	mde_mil_consistency_weight=2	0.851	0.865	0.849
WSI bag	Balanced MIL	sampler_power=0.75	0.821	0.831	0.829
WSI bag	Focal MIL	focal_gamma=0.5	0.823	0.824	0.823
WSI bag	Weighted MIL	weight_power=0.125	0.828	0.835	0.833
WSI bag	RankMix	rankmix_alpha=32	0.843	0.854	0.864
WSI bag	SC-MIL	sc_mil_temperature=0.1	0.828	0.839	0.837

Table 6: Validation-selected configuration provenance. Configurations are chosen by mean validation macro F1 over seeds 0–2; held-out test metrics in Tables 7 and 9 are reported after selection.

twice the seed standard deviation only for OKO (range-to-seed-noise ratio 75.8) and RankMix (ratio 2.06, $p < 0.01$). For all single-knob resampling, reweighting, and focal mechanisms on frozen Virchow2 embeddings the swept control produces variation within seed noise regardless of its value; the selected configurations in Table 6 therefore reflect the best observed point rather than a robustly isolated tuning effect.

5.2 Patch-Level Results

Method	Accuracy	Balanced accuracy	Macro F1
CFAL	0.847 ± 0.008	0.789 ± 0.007	0.798 ± 0.009
OKO	0.841 ± 0.006	0.791 ± 0.005	0.788 ± 0.003
CE + soft F1 (balanced)	0.835 ± 0.004	0.781 ± 0.002	0.780 ± 0.003
Weighted CE	0.826 ± 0.008	0.767 ± 0.006	0.774 ± 0.005
CE	0.828 ± 0.010	0.764 ± 0.011	0.773 ± 0.015
Balanced sampler	0.827 ± 0.010	0.769 ± 0.007	0.771 ± 0.014
CE + soft MCC (balanced)	0.824 ± 0.009	0.772 ± 0.004	0.770 ± 0.005
ProGAN augmentation	0.824 ± 0.004	0.760 ± 0.005	0.766 ± 0.002
Focal	0.825 ± 0.007	0.758 ± 0.007	0.765 ± 0.009

Table 7: Validation-selected patch-level test results over three seeds. Methods share the same controlled manifest, frozen Virchow2 encoder, shared MLP trunk, and evaluation code.

The patch benchmark compresses into a narrow performance band once patches are represented with frozen Virchow2 embeddings (Table 7). CFAL ranks highest on macro F1, while OKO ranks highest on balanced accuracy. CFAL has paired gains over plain CE on both headline metrics in all three seeds (Table 12). OKO edges CFAL on balanced accuracy and stays above CE plus soft F1 on that metric, while achieving a macro F1 between CE plus soft F1 and CE plus soft MCC. CE plus soft F1 and OKO both improve on CE in the paired comparison, but the absolute gaps among these methods remain modest and standard deviations overlap in the summary table. Weighted CE, balanced sampling, and focal loss remain within the broader CE-centered band rather than separating cleanly from it.

The simpler patch interventions separate mainly by how little they move the validation-selected classifier after freezing Virchow2. Class-weighted CE, balanced sampling, focal loss, and ProGAN augmentation remain near CE on mean macro F1. ProGAN augmentation stays within the non-generative spread despite adding a large synthetic tail-class volume (Table 8). One plausible explanation is the frozen-feature pipeline rather than poor generator quality. The class-specific generators reach an Inception FID of 239.5–382.0 (median 293.4), which is comparable to the ProGAN FID of 222–261 reported by Ruiz-Casado et al. on a binary breast-histology task where GAN augmentation still produced large downstream gains (Ruiz-Casado et al., 2024). The decisive difference is how the synthetic patches enter the classifier: the source method trains a convolutional classifier end to end on pixels, so it can adapt its features to the generated distribution, whereas this benchmark embeds every patch with a frozen Virchow2 encoder and trains only a lightweight head. A frozen encoder may map synthetic patches that are slightly off the real distribution into embedding regions the head rarely sees for real data, which would limit the value of additional synthetic volume even when pixel-level FID is reasonable. The per-class FID and Virchow2 nearest-neighbor distances in Table 8 are therefore reported as descriptive quality indicators rather than as proof of the null mechanism; the nearest-neighbor distances in particular lack a real-to-real baseline and should not be read as direct evidence of off-manifold generation. This frozen-feature explanation remains a hypothesis unless it is supported by feature-space real-versus-synthetic diagnostics such as real-to-real nearest-neighbor baselines, maximum mean discrepancy, or kNN class-mixing analysis. The small focal-loss shift is consistent with the interpretation that difficulty reweighting is not very helpful when most cancer types are already separable in the frozen embedding space.

Class	Real train	Generated	Inception FID	Virchow2 mean NN
Lymphoid Neoplasm Diffuse Large B-cell Lymphoma	600	16,240	345.6	50.68
Cholangiocarcinoma	660	16,180	307.6	51.79
Pheochromocytoma and Paraganglioma	900	15,940	382.0	54.06
Uveal Melanoma	1,160	15,680	338.4	48.88
Rectum adenocarcinoma	1,350	15,490	239.5	47.97
Mesothelioma	1,430	15,410	286.3	51.23
Uterine Carcinosarcoma	1,550	15,290	304.0	50.94
Kidney Chromophobe	1,740	15,100	282.8	48.97
All augmented classes (31 total)	165,470	356,570	293.4 median	51.32 median

Table 8: ProGAN augmentation diagnostics for seed 0. "Generated" counts are synthetic patches added per class to reach the head-class training patch count. Inception FID compares generated patches to the real training subset used for GAN training. Virchow2 mean NN is the mean, over a fixed synthetic subsample, of the minimum L_2 distance to real same-class embeddings in the frozen feature cache.

The classwise-recall plot (Figure 5) shows that observed errors are distributed across many cancer types rather than concentrated in a single tail class. In this setting, the main practical distinction is therefore not whether imbalance mitigation works at all, but which mechanism best preserves rare-class recall when the feature extractor is already strong. These conclusions are confined to the patch benchmark: high patch accuracy or macro F1 does not directly establish slide-level utility, because each test prediction is made on a single crop rather than by aggregating evidence across a slide bag.

5.3 WSI-Bag Results

Method	Accuracy	Balanced accuracy	Macro F1
MDE-MIL (ensemble)	0.896 ± 0.002	0.849 ± 0.014	0.865 ± 0.011
RankMix	0.890 ± 0.005	0.864 ± 0.002	0.854 ± 0.008
SC-MIL	0.883 ± 0.013	0.837 ± 0.013	0.839 ± 0.008
Weighted MIL	0.882 ± 0.006	0.833 ± 0.019	0.835 ± 0.011
Balanced MIL	0.875 ± 0.003	0.829 ± 0.010	0.831 ± 0.009
MIL CE	0.873 ± 0.013	0.823 ± 0.014	0.826 ± 0.013
Focal MIL	0.872 ± 0.006	0.823 ± 0.019	0.824 ± 0.008

Table 9: Validation-selected WSI-bag test results over three seeds. Methods share the same capped feature bags, attention-MIL backbone family, and evaluation code.

The WSI-bag benchmark evaluates one slide-level prediction per diagnostic slide by aggregating capped feature bags from the shared split. The 30-instance cap aligns the per-slide evidence budget with the controlled patch manifest and prevents larger slides from benefiting simply by contributing more instances. The benchmark difference is architectural and supervisory: MIL learns slide-level attention over a set of instances, whereas the patch classifier treats each crop as an independently labeled example. Table 9 should be read as a within-regime slide-level leaderboard, not as confirmation or refutation of the patch-level ranking.

RankMix is above plain MIL CE on both headline metrics in all three paired seeds (Table 12) and has the highest mean balanced accuracy in this benchmark, although its macro-F1 gain is more modest than its balanced-accuracy gain and seed-wise standard deviations overlap other methods (Table 9, Figure 6). MDE-MIL reaches the highest mean test macro F1, with positive paired macro-F1 deltas in all three seeds but a wider spread in absolute macro F1 than RankMix. On raw probability quality, MDE-MIL has the lowest mean negative log likelihood (NLL), Brier score, and expected calibration error (ECE) among WSI methods, with RankMix second; both clearly improve on plain MIL CE before temperature scaling.

Other WSI methods cluster closely below this pair. Balanced-sampling MIL and SC-MIL trail RankMix and MDE-MIL in macro F1. SC-MIL performs well on validation but does not exceed plain MIL CE on the test macro-F1 average. Weighted MIL CE and focal MIL do not consistently improve over plain MIL CE. Focal MIL is in particular unstable across seeds: it is competitive with the stronger WSI methods on seeds 0 and 2 (test macro F1 0.846 and 0.845) but collapses on seed 1 (0.811), the single lowest per-seed point in the benchmark (Figure 6). Its seed-averaged macro F1 (0.834) is therefore driven by one low-outlier run rather than a uniform deficit, and its seed-to-seed standard deviation (0.020) is the widest among the WSI methods. Activation diagnostics confirm that the advanced WSI objectives were active: RankMix generated about 1,100,000 mixed examples per seed, SC-MIL constructed about 1,060,000 positive pairs per seed, and MDE-MIL processed both natural and balanced branches every training step.

5.4 Paired Seed Comparisons

Table 12 makes the headline gains explicit at seed resolution rather than only through aggregate means. CFAL is above plain CE on macro F1 and balanced accuracy in every seed, with larger shifts than the Scholz-style hybrids. CE plus soft F1 is above plain CE on balanced accuracy in every seed and on macro F1 in two of the three seeds, while CE plus soft MCC is positive on balanced accuracy in every seed but has one slightly negative macro-F1 delta. OKO is above plain CE on both headline metrics in every seed, with balanced accuracy gains comparable to CFAL and macro F1 gains between CE plus soft F1 and CE plus soft MCC. On WSI bags, both RankMix and MDE-MIL are above plain MIL CE on macro F1 and balanced accuracy in every seed. The absolute shifts are modest and would require formal paired testing to treat as significant.

5.5 Probability Calibration

Table 13 applies the metrics in Section 4.1 to the validation-selected configurations and makes the discrimination–calibration tradeoff explicit for raw outputs. On patches, the method with the lowest ECE is not the method with the strongest macro F1, and the Scholz-style hybrid objectives improve class-balanced discrimination without uniformly improving calibration. CFAL improves discrimination by the largest margin among patch methods but has the highest raw ECE and the worst NLL/Brier scores in Table 13. CFAL's reported confidences are systematically too high relative to empirical accuracy—a typical over-confidence pattern when geometric affinities are passed through $\log a_{ic}$ and softmax—so it should be read as a strong patch-level *discriminator*, not as a calibrated probabilistic classifier on its raw outputs. OKO is the notable exception to the usual discrimination–calibration tradeoff on likelihood-based scores: it achieves the best raw NLL and Brier among patch methods while also placing highest on balanced accuracy. Its raw ECE, however, is not lowest: focal loss attains the smallest raw ECE, so no patch method dominates all three raw calibration metrics. The implicit logit regularization from averaging per-example outputs across set members limits how peaked the aggregate logit vector can become, which appears to prevent the overconfidence that afflicts CFAL and many CE variants. ProGAN augmentation further weakens probability quality relative to the CE baseline. On WSI bags, MDE-MIL and RankMix both raise mean balanced accuracy and sharply reduce raw ECE relative to plain MIL CE; MDE-MIL reaches the lowest raw NLL, Brier, and ECE, with RankMix close behind, while the other MIL variants cluster near the MIL CE baseline on all three raw metrics.

Regime	Method	NLL	Brier	ECE
Patch	CFAL	0.532 ± 0.029	0.220 ± 0.011	0.019 ± 0.003
Patch	OKO	0.617 ± 0.032	0.236 ± 0.011	0.029 ± 0.006
Patch	CE + soft F1 (balanced)	0.780 ± 0.026	0.263 ± 0.008	0.061 ± 0.028
Patch	Weighted CE	0.654 ± 0.030	0.252 ± 0.012	0.038 ± 0.005
Patch	CE	0.649 ± 0.035	0.250 ± 0.014	0.041 ± 0.002
Patch	Balanced sampler	0.653 ± 0.034	0.252 ± 0.013	0.038 ± 0.008
Patch	CE + soft MCC (balanced)	0.768 ± 0.034	0.271 ± 0.013	0.076 ± 0.029
Patch	ProGAN augmentation	0.723 ± 0.017	0.263 ± 0.005	0.073 ± 0.007
Patch	Focal	0.618 ± 0.025	0.251 ± 0.009	0.006 ± 0.004
WSI bag	MDE-MIL (ensemble)	0.471 ± 0.009	0.163 ± 0.003	0.023 ± 0.002
WSI bag	RankMix	0.428 ± 0.012	0.166 ± 0.005	0.028 ± 0.002
WSI bag	SC-MIL	0.440 ± 0.045	0.173 ± 0.018	0.007 ± 0.004
WSI bag	Weighted MIL	0.455 ± 0.033	0.177 ± 0.011	0.013 ± 0.004
WSI bag	Balanced MIL	0.479 ± 0.031	0.185 ± 0.010	0.013 ± 0.003
WSI bag	MIL CE	0.473 ± 0.018	0.188 ± 0.015	0.017 ± 0.010
WSI bag	Focal MIL	0.475 ± 0.024	0.189 ± 0.008	0.012 ± 0.002

Table 10: Held-out test calibration metrics for the validation-selected configurations after method-agnostic post-hoc temperature scaling. For each method and seed, the temperature is fit only on that seed's validation split and then applied to the test split. This table is the decision-facing probability comparison; Table 13 is the raw-score diagnostic.

Table 10 reports post-hoc NLL/Brier/ECE after the shared temperature-scaling protocol. On patches, temperature scaling largely removes the raw-score inversion between CFAL and CE. CFAL reaches the best mean NLL and Brier score, while focal loss reaches the lowest mean post-hoc ECE. This is consistent with treating affinity outputs as rank scores that need a monotone probability map rather than as intrinsically miscalibrated classifiers. On WSI bags, RankMix has the lowest mean post-hoc NLL and MDE-MIL the lowest mean post-hoc Brier, with the two methods close on both. After scaling, however, the simpler MIL baselines—SC-MIL, focal MIL, and plain MIL CE—reach lower mean ECE than either RankMix or MDE-MIL, so the raw RankMix and MDE-MIL ECE advantage over MIL CE should be read partly as a confidence-scale mismatch rather than as proof that they remain uniformly better calibrated after a shared post-hoc step.

All WSI-bag calibration scores are computed from the same held-out test slides, hard slide labels, and raw softmax outputs of the final-epoch model. An audit over stored test-result files confirms that all seven WSI methods share identical test label sequences within each seed, that reported probabilities sum to one, and that stored NLL, Brier, and ECE values match recomputation from the saved probability matrix.

5.6 Classwise Difficulty and Support

The classwise recall analysis shows that errors are uneven across cancer types, but support tiers alone do not reveal whether rare classes are always the hard classes. A post-hoc difficulty analysis therefore uses the CE baseline in each regime as the reference error pattern and compares each class's full-dataset slide support with its seed-averaged held-out recall. Difficulty is operationalized as the false-negative rate, $1 - \text{recall}$, so the analysis asks whether classes with lower slide support are also the classes with higher baseline error.

Regime	Class	Slides	Support rank	Difficulty rank	Recall	F1
Patch	Rectum adenocarcinoma	63	5	1	0.103	0.144
Patch	Cholangiocarcinoma	30	2	2	0.319	0.409
Patch	Esophageal carcinoma	113	10	3	0.362	0.420
Patch	Uterine Carcinosarcoma	71	7	4	0.539	0.519
Patch	Mesothelioma	70	6	5	0.556	0.628
WSI-bag	Rectum adenocarcinoma	63	5	1	0.338	0.409
WSI-bag	Esophageal carcinoma	113	10	2	0.523	0.564
WSI-bag	Cholangiocarcinoma	30	2	3	0.528	0.544
WSI-bag	Uterine Carcinosarcoma	71	7	4	0.625	0.611
WSI-bag	Stomach adenocarcinoma	274	18	5	0.693	0.735

Table 11: Hardest classes under each regime's CE baseline, averaged over seeds 0–2 on the held-out test split. Support rank is computed from full TCGA-UT slide counts, with rank 1 indicating the lowest-support class. Difficulty rank orders classes by baseline error rate, $1 - \text{recall}$.

Table 11 and Figures 8–9 make the support–difficulty mismatch explicit. Rectum adenocarcinoma is the hardest class in both regimes despite being the fifth-lowest-support class rather than the rarest class. Conversely, the rarest class, diffuse large B-cell lymphoma, does not appear among the five hardest classes in either regime. The rank association between dataset slide support and CE-baseline recall is positive but modest: Spearman $\rho = 0.334$ for patches and $\rho = 0.301$ for WSI bags. These descriptive correlations support a constrained interpretation: class frequency contributes to difficulty, but it does not fully determine which class boundaries remain hard in the frozen Virchow2 feature space.

6 Discussion

The two benchmarks support different but compatible conclusions. In the patch benchmark, a strong frozen encoder compresses most methods into a narrow band, but CFAL stands out most by replacing the linear head with prototype-based affinity training. OKO provides consistent gains on balanced accuracy comparable to CFAL and on macro F1 comparable to the Scholz-style hybrids, and it combines near-best discrimination with the best raw NLL and Brier among patch methods, although focal loss has lower raw ECE. The implicit mechanism is logit averaging across set members: each set prediction is the sum of individual forward passes, which limits how peaked the aggregated logit can become and thereby suppresses the overconfidence that affects CE-trained classifiers. The Scholz-style hybrid objectives provide modest improvements over plain CE rather than large separations, and the simpler reweighting or resampling variants remain within the broader CE overlap band. ProGAN augmentation does not deliver a clear advantage even though it adds large numbers of synthetic tail-class patches; one plausible explanation is the frozen-feature pipeline, since the generators reach an Inception FID comparable to the source study that reported large gains, and that study adapted its classifier to the synthetic images end to end whereas this benchmark consumes them through a frozen encoder and a lightweight head. That explanation remains a hypothesis unless feature-space real-versus-synthetic diagnostics confirm it. Focal loss is also unhelpful here, consistent with the interpretation that difficulty reweighting is redundant when the representation has already absorbed much of the class-separation problem. In the WSI-bag benchmark, RankMix has the highest mean balanced accuracy and is above MIL CE on balanced accuracy in every seed, while MDE-MIL reaches the highest mean macro F1 and the best raw probability scores, though with higher seed variance; these WSI gaps are small relative to the seed-to-seed spread.

These readings should be interpreted within the benchmark's deliberate scope. The experiment is controlled rather than a full reproduction of every original paper: method sections document the necessary adaptations to frozen Virchow2 features and TCGA-UT supervision, and both regimes use frozen features rather than end-to-end encoder adaptation. Each non-baseline method is tuned over one imbalance-specific control on validation; the search is compact rather than an exhaustive optimization of every published implementation. The resulting rankings therefore describe how imbalance mechanisms behave when patch and slide classification are already supported by a strong pathology foundation model.

The dataset's moderate imbalance also means that class frequency is only one part of the difficulty signal. The classwise difficulty analysis shows this directly: several low-support classes are hard, but the hardest classes are not exactly the rarest classes, and support explains only a modest share of the classwise recall ordering. This distinction helps explain why pure exposure corrections have limited room to improve the patch benchmark. Balanced sampling and class-weighted CE increase the influence of observed minority examples, but they do not add new within-class diversity or change the separability of visually related classes in the Virchow2 feature space.

Holding the encoder fixed within each benchmark therefore changes what an imbalance intervention can accomplish. On patches, the bottleneck shifts from learning stain and tissue morphology to calibrating decision boundaries under an imbalanced label distribution. Under that bottleneck, CFAL's prototype shaping provides the largest discrimination gain, whereas objectives that reshape training at the set or minibatch level—OKO through set-aggregated logit regularization, and the Scholz-style hybrids through metric-derived loss terms—provide more moderate discrimination improvements; OKO additionally yields the best raw NLL and Brier among patch methods. On slides, the bottleneck instead includes instance selection and bag aggregation, so RankMix can add value by mixing informative subsequences even when the underlying features are strong. MDE-MIL shows that coupling natural and balanced branches with cross-expert consistency can raise mean macro F1 and gives the best raw probability scores, although the macro-F1 gain is seed-sensitive and RankMix has the highest mean balanced accuracy in this benchmark.

These results should be read within each benchmark's prediction unit. Patch accuracy does not directly imply slide-level deployment quality, and WSI-bag gains may reflect aggregation behavior absent from the patch classifier. The shared split, encoder family, and 30-instance slide budget link the two benchmarks descriptively, but the evidence for each method comes from the table corresponding to its evaluated input.

Several limitations apply. TCGA-UT patch labels come from pathologist-selected tumor regions; they are useful patch-level class labels but not dense annotations. The two benchmarks share case-disjoint splits but differ in prediction unit and per-slide evidence budget, so their result tables are non-transferable. The validation sweep varies one imbalance-specific control per method and does not exhaust architecture, training schedule, multiple interacting hyperparameters, or remaining ProGAN generator-side choices beyond the final-depth refinement epochs (latent dimension, base channels, and learning rate). In particular, ProGAN augmentation is evaluated only through the frozen Virchow2 pipeline; whether GAN-generated patches help under end-to-end encoder adaptation is left open and is the most plausible route to a non-null synthetic-augmentation result on this dataset.

7 Conclusion

TCGA-UT provides a native, moderately imbalanced setting representative of real-world clinical imbalance for comparing representative imbalance interventions on frozen Virchow2 patch and WSI-bag representations. After validation-selected tuning on case-disjoint splits, CFAL reaches the highest patch-level mean macro F1; OKO set learning reaches the highest balanced accuracy while achieving the best raw NLL and Brier among patch methods; and CE plus soft F1 with balanced sampling also improves both discrimination metrics, but with narrower gaps. No patch method is uniformly best across raw and post-hoc calibration metrics. ProGAN augmentation and focal loss remain close to the CE baseline. On WSI bags, RankMix has the highest mean balanced accuracy and MDE-MIL the highest mean macro F1, with MDE-MIL also giving the best raw probability scores (lowest NLL, Brier, and ECE); these WSI gaps are small relative to seed variation, so they should be read as the highest means in this benchmark rather than as established differences. The practical implication is that imbalance mitigation should be evaluated in the native input regime and alongside classwise recall and calibration, not only by aggregate discrimination.

A Additional Result Tables and Figures

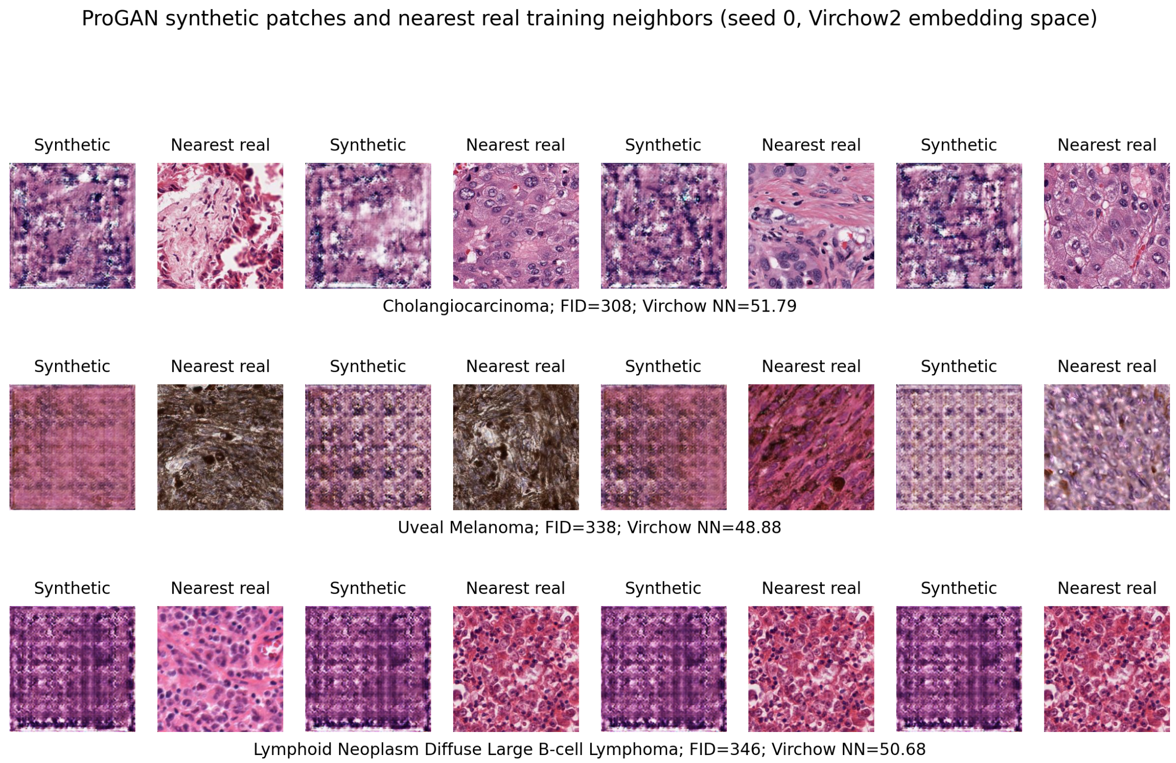


Figure 4: ProGAN synthetic patches (left column of each pair) and their nearest real training neighbors in Virchow2 embedding space (right column) for three tail classes in seed 0. Visual quality is class-dependent: the Cholangiocarcinoma generator still produces clear color-banding artifacts, whereas the Uveal Melanoma and Lymphoid Neoplasm generators yield patches that look closer to real histology. Visual quality is therefore not the limiting factor: the per-class Inception FID values are comparable to those reported for ProGAN in the source augmentation study, and ProGAN augmentation is interpreted through the frozen-feature pipeline—synthetic patches are embedded by a frozen Virchow2 encoder and consumed by a lightweight head—rather than through pixel-level appearance or FID alone.

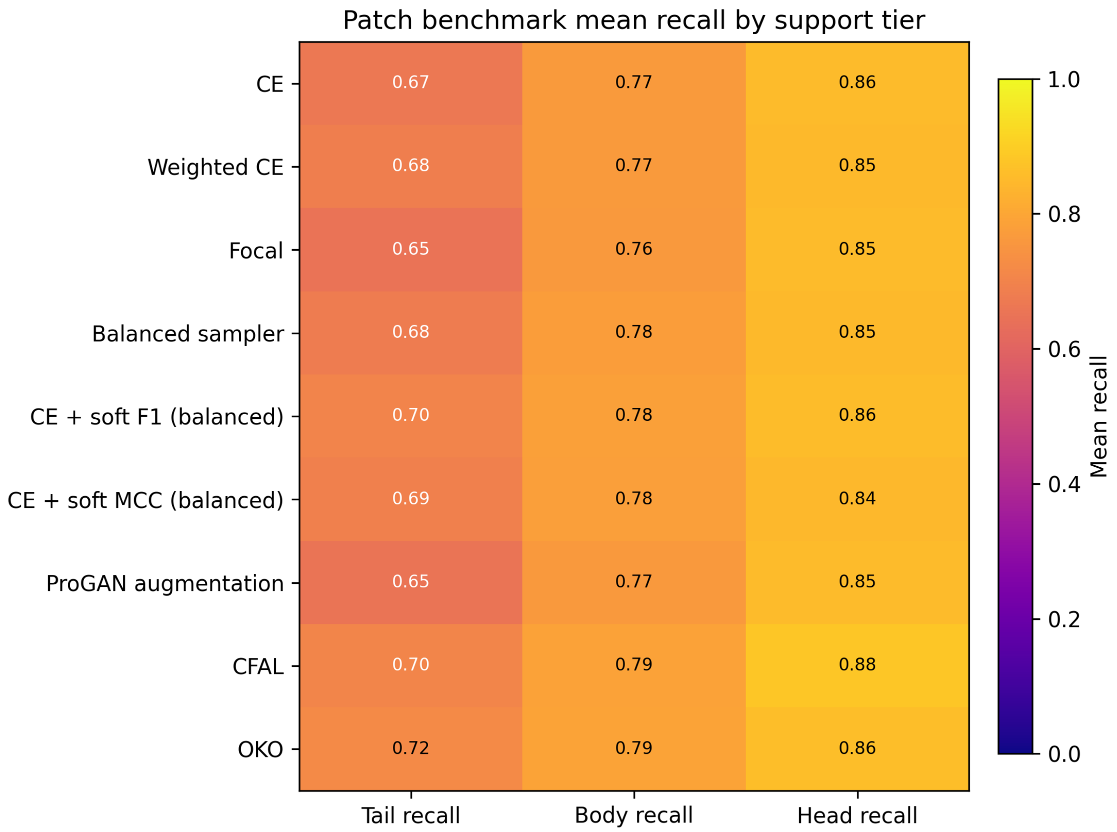


Figure 5: Patch-level mean recall by support tier on the held-out test split. The heatmap exposes the head–body–tail variation hidden by aggregate accuracy and motivates reporting balanced accuracy and macro F1.

Comparison	Δ Macro F1	Δ Balanced accuracy
CE + soft MCC − CE	-0.003 ± 0.008 $[+0.005, -0.000, -0.014]$	$+0.009 \pm 0.007$ $[+0.015, +0.012, -0.000]$
CE + soft F1 − CE	$+0.007 \pm 0.010$ $[+0.014, +0.014, -0.007]$	$+0.018 \pm 0.008$ $[+0.023, +0.023, +0.007]$
OKO − CE	$+0.015 \pm 0.010$ $[+0.024, +0.021, +0.001]$	$+0.028 \pm 0.012$ $[+0.038, +0.033, +0.011]$
CFAL − CE	$+0.025 \pm 0.005$ $[+0.030, +0.027, +0.019]$	$+0.026 \pm 0.004$ $[+0.030, +0.027, +0.020]$
RankMix − MIL CE	$+0.028 \pm 0.010$ $[+0.023, +0.041, +0.020]$	$+0.041 \pm 0.010$ $[+0.038, +0.055, +0.031]$
MDE-MIL − MIL CE	$+0.039 \pm 0.016$ $[+0.055, +0.043, +0.018]$	$+0.026 \pm 0.016$ $[+0.041, +0.034, +0.003]$

Table 12: Paired validation-selected test-split differences against the within-regime CE baseline on seeds 0–2. Each cell reports mean $\pm$ SD across seeds; bracketed values list seed-ordered paired deltas. No formal p-values are computed.

Regime	Method	NLL	Brier	ECE
Patch	CFAL	2.850 ± 0.011	0.914 ± 0.001	0.787 ± 0.008
Patch	OKO	0.756 ± 0.040	0.270 ± 0.015	0.158 ± 0.013
Patch	CE + soft F1 (balanced)	3.038 ± 0.137	0.316 ± 0.008	0.155 ± 0.004
Patch	Weighted CE	1.965 ± 0.115	0.314 ± 0.016	0.147 ± 0.007
Patch	CE	1.976 ± 0.149	0.312 ± 0.018	0.147 ± 0.009
Patch	Balanced sampler	1.967 ± 0.139	0.313 ± 0.019	0.147 ± 0.009
Patch	CE + soft MCC (balanced)	2.870 ± 0.153	0.332 ± 0.017	0.162 ± 0.008
Patch	ProGAN augmentation	2.511 ± 0.093	0.327 ± 0.007	0.157 ± 0.004
Patch	Focal	1.370 ± 0.081	0.298 ± 0.014	0.131 ± 0.006
WSI bag	MDE-MIL (ensemble)	0.487 ± 0.005	0.166 ± 0.002	0.036 ± 0.004
WSI bag	RankMix	0.584 ± 0.029	0.170 ± 0.006	0.049 ± 0.001
WSI bag	SC-MIL	1.024 ± 0.145	0.204 ± 0.020	0.092 ± 0.009
WSI bag	Weighted MIL	1.224 ± 0.144	0.212 ± 0.012	0.099 ± 0.006
WSI bag	Balanced MIL	1.264 ± 0.165	0.223 ± 0.010	0.102 ± 0.005
WSI bag	MIL CE	1.283 ± 0.080	0.227 ± 0.024	0.106 ± 0.012
WSI bag	Focal MIL	1.117 ± 0.153	0.221 ± 0.014	0.098 ± 0.009

Table 13: Held-out test calibration metrics for the validation-selected configurations over three seeds from raw final-epoch probabilities. Lower is better for negative log likelihood (NLL), multiclass Brier score, and expected calibration error (ECE). Values are not comparable across regimes because patch and WSI-bag models operate on different input units.

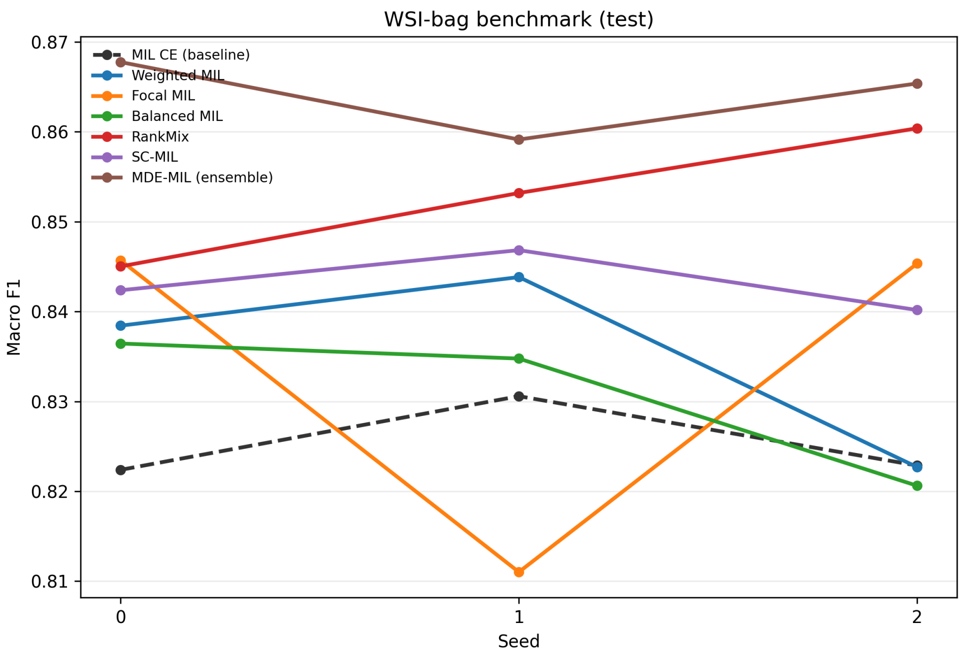


Figure 6: Validation-selected WSI-bag macro F1 on the held-out test split by random seed. The dashed line marks plain MIL CE; seed-wise spread shows why cross-method gaps should be read cautiously when standard deviations overlap (Table 9). The pronounced focal-MIL dip on seed 1 is a single-seed training instability: focal MIL tracks the strongest methods on seeds 0 and 2 but collapses on seed 1, so its averaged result reflects one unstable run rather than a consistent deficit.

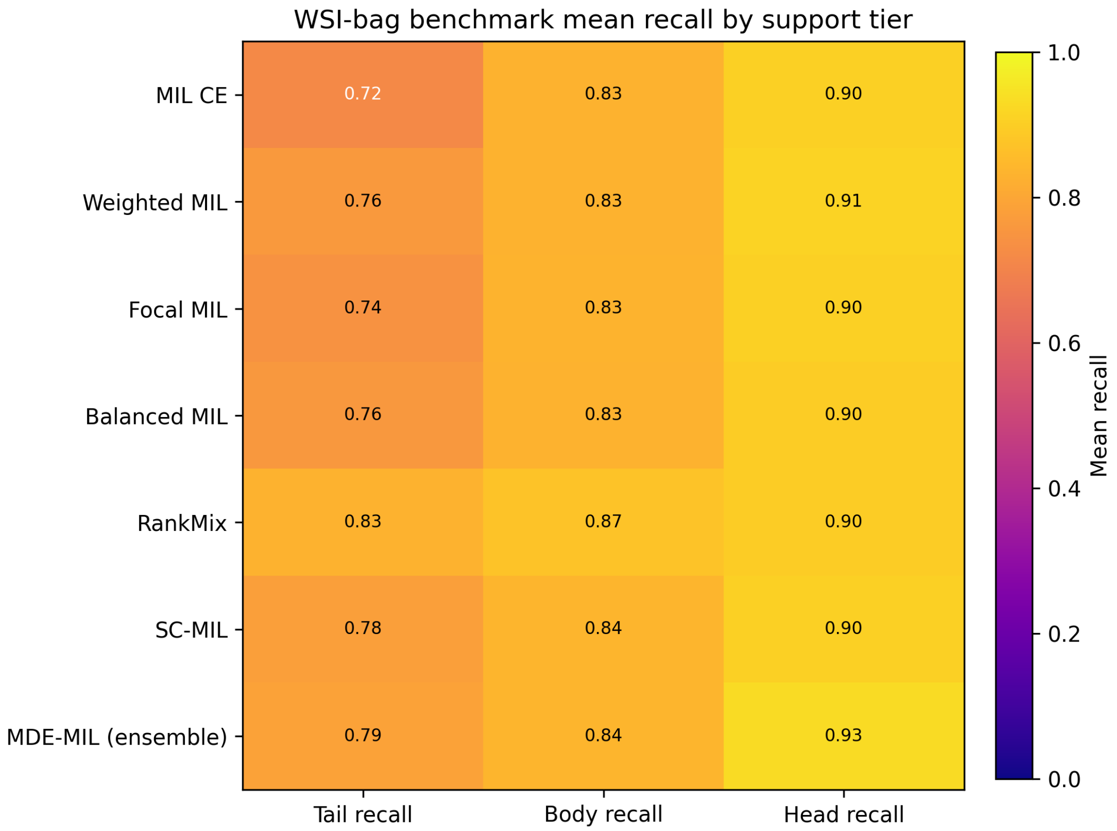


Figure 7: WSI-bag classwise recall on the held-out test split. The spread across classes shows why aggregate accuracy alone is insufficient for the imbalanced setting.

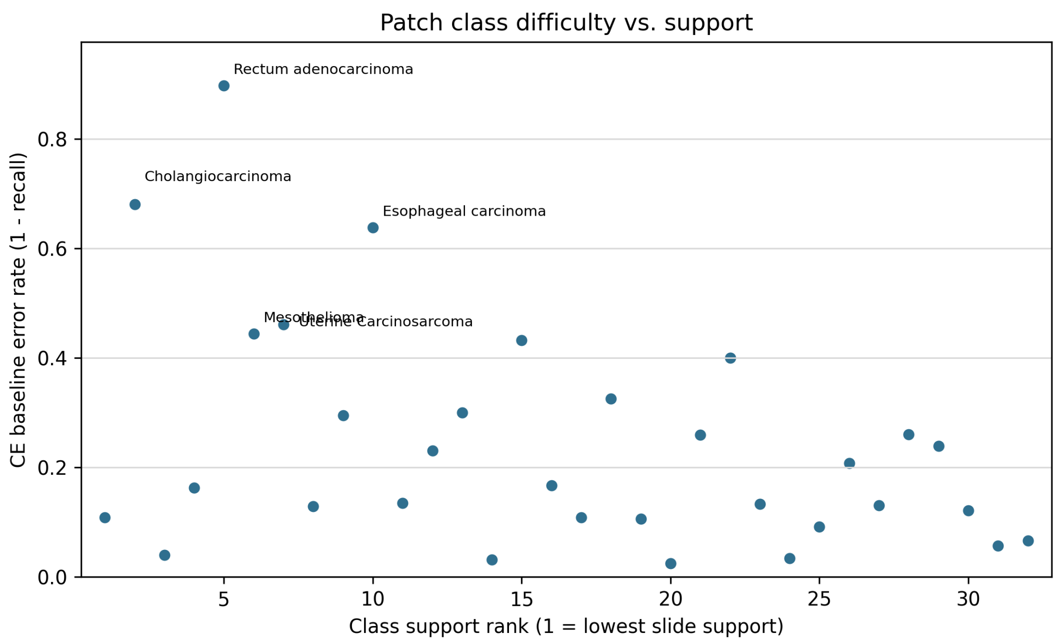


Figure 8: Patch CE-baseline class error rate versus full-dataset class-support rank. The hardest patch classes are not simply ordered by rarity, indicating that exposure-based corrections address only part of the error pattern.

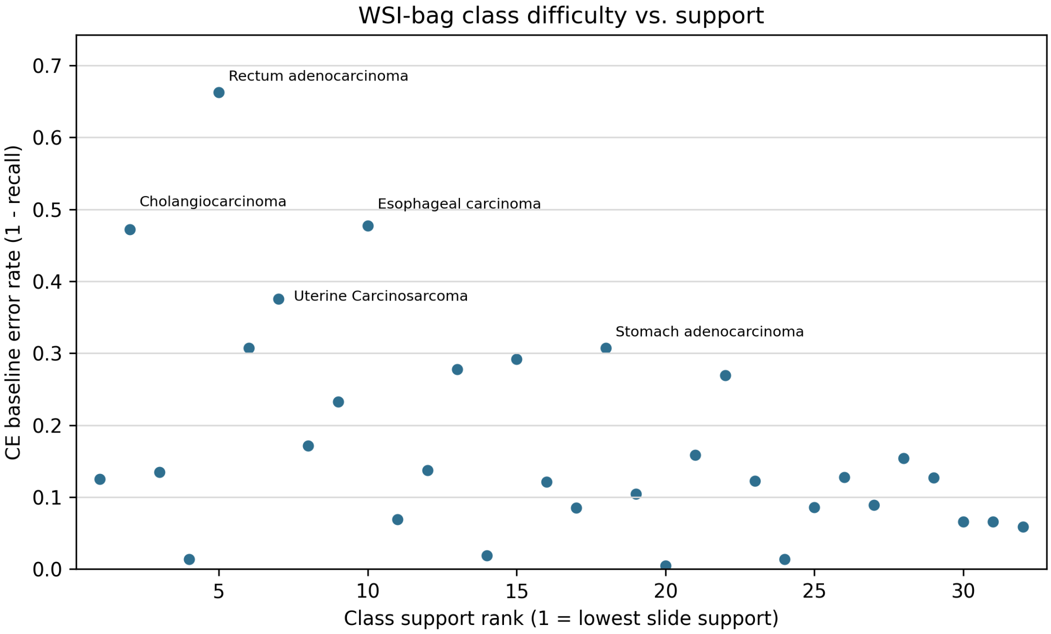


Figure 9: WSI-bag CE-baseline class error rate versus full-dataset class-support rank. The same high-error classes are visible at slide level, showing that class difficulty is not reducible to class frequency.

B Terminology

Term	Meaning in this study
Slide / WSI	Diagnostic whole-slide image carrying one cancer-type label.
Case / patient	TCGA participant identified by the first three fields of the TCGA slide barcode, such as `TCGA-OR-A5J1`.
Patch	256×256 H&E crop from a selected tumor region; patch labels inherit the slide cancer type.
Patch embedding	Frozen Virchow2 vector for one controlled patch; input to the patch-level classifier head.
Region	Tumor area selected before patch extraction.
Cancer type	The 32-way class label used for patch and slide-level classification.
Feature chunk	Frozen Virchow2 feature representation used to assemble slide-level feature bags.
WSI bag	Set of patch-feature instances for one slide, consumed by the MIL model.
Head/body/tail class	Fixed support tier from the full TCGA-UT slide distribution; see Section 3.8.
Case-disjoint split	Split rule preventing any TCGA participant's slides, patches, or features from crossing train, validation, and test.

Table 14: Terminology used for the two native input regimes (see Section 2.1).

References

Brier, G. W. (1950). Verification of forecasts expressed in terms of probabilities. *Monthly Weather Review*, 78(1):1–3.

Chen, Y.-C. and Lu, C.-S. (2023). Rankmix: Data augmentation for weakly supervised learning of classifying whole slide images with diverse sizes and imbalanced categories. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 23936–23945.

Guo, C., Pleiss, G., Sun, Y., and Weinberger, K. Q. (2017). On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning*, pages 1321–1330. PMLR.

Juyal, D., Shingi, S., Javed, S. A., Padigela, H., Shah, C., Sampat, A., Khosla, A., Abel, J., and Taylor-Weiner, A. (2024). SC-MIL: Supervised contrastive multiple instance learning for imbalanced classification in pathology. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, pages 7016–7025.

Komura, D., Kawabe, A., Fukuta, K., Sano, K., Umezaki, T., Koda, H., Suzuki, R., Tominaga, K., Ochi, M., Konishi, H., et al. (2022). Universal encoding of pan-cancer histology by deep texture representations. *Cell Reports*, 38(9):110424.

Ling, X., Ping, Y., Li, J., Peng, J., and Chen, Y. (2025). Mde-mil. *Dos mil tres mil*, 27.

Mahbub, T., Obeid, A., Javed, S., and Dias, J. (2024). Center-focused affinity loss for class imbalance histology image classification. *IEEE Journal of Biomedical and Health Informatics*, 28(2):952–963.

Mosquera, C., Ferrer, L., Milone, D. H., Luna, D., and Ferrante, E. (2024). Class imbalance on medical image classification: towards better evaluation practices for discrimination and calibration performance. *European Radiology*, 34(12):7895–7903.

Muttenthaler, L., Vandermeulen, R. A., Zhang, Q., Unterthiner, T., and Müller, K.-R. (2024). Set learning for accurate and calibrated models. In *Proceedings of the Twelfth International Conference on Learning Representations*.

Naeini, M. P., Cooper, G., and Hauskrecht, M. (2015). Obtaining well calibrated probabilities using Bayesian binning. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 29, pages 2901–2907.

Ruiz-Casado, J. L., Molina-Cabello, M. A., and Luque-Baena, M. R. (2024). Enhancing histopathological image classification performance through synthetic data generation with generative adversarial networks. *Sensors*, 24(12):3777.

Saini, M. and Susan, S. (2023). Tackling class imbalance in computer vision: a contemporary review. *Artificial Intelligence Review*, 56(S1):1279–1335.

Scholz, D., Erdur, A. C., Buchner, J., Peeken, J. C., Rueckert, D., and Wiestler, B. (2024). Imbalance-aware loss functions improve medical image classification. In *Proceedings of the 7th International Conference on Medical Imaging with Deep Learning*, volume 250 of *Proceedings of Machine Learning Research*, pages 1341–1356. PMLR.

Tomczak, K., Czerwinska, P., and Wiznerowicz, M. (2015). The cancer genome atlas (tcga): an immeasurable source of knowledge. *Contemporary Oncology*, 19(1A):A68–A77.

Vorontsov, E., Bozkurt, A., Casson, A., Shaikovski, G., Zelechowski, M., Severson, K., Zimmermann, E., et al. (2024). A foundation model for clinical-grade computational pathology and rare cancers detection. *Nature Medicine*, 30(10):2924–2935.

Zhang, Y., Kang, B., Hooi, B., and Yan, S. (2023). Deep long-tailed learning: A survey. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(9):10795–10816.

Zimmermann, E., Vorontsov, E., Viret, J., Casson, A., Zelechowski, M., and Shaikovski, G. (2024). Virchow2. *2024 Shell TechXplorer Digest*, 2024(1).